HO CHI MINH CITY UNIVERSITY OF TRANSPORT (UTH)
FACULTY OF INFORMATION TECHNOLOGY

Capstone Project Document

Horse Racing Tournament Management System

Group Members	Ha Duy Hung - 074206002060 *(Team Leader)* Dinh Thuy Anh - 068306011497 Ho Nguyen Thai Chau - 054206003685 Le Thi My Hue - 052306012275 Bui Le Quang Huy - 231206000064 Nguyen Gia Huy - 2251120017 Doan Thi Thu May - 068306004742
Supervisor	Lecturer Nguyen Van Chien
Capstone Project code	Hourse Racing Tournament

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We would also like to thank every member of our team for their effort and close collaboration, which made it possible to complete the "Horse Racing Tournament Management System" project on schedule.

Definition and Acronyms

Term / Abbreviation	Definition
MVP	Minimum Viable Product
SRS	Software Requirement Specification
SDD	Software Design Description
ERD	Entity Relationship Diagram
UC	Use Case
BR	Business Rule
CR	Common Requirement
MSG	Application Message
UI	User Interface
API	Application Programming Interface
DB	Database

I. Project Introduction

1.1. Overview

1.1.1 Project Information

- **Project name:** Horse Racing Tournament Management System
- **Project code:**
- **Group name:**
- **Software Type:** Web application

1.1.2 Project Team

Full Name	Role	Email	Mobile
	Supervisor		
	Leader		
	Member		
	Member		
	Member		

1.2. Product Background

Write product background here.

1.3. Existing Systems

1.3.1 Existing System Name

Description: Write description here.
Drawbacks:

- Drawback 1
- Drawback 2
- Drawback 3

1.4. Business Opportunity

Write business opportunity here.

1.5. Software Product Vision

Write product vision here.

1.6. Project Scope & Limitations

1.6.1 Major Features

Major Features for Admin

FE-01: Login/Logout
FE-02: Manage user accounts

Major Features for Horse Owner

FE-01: Register horse for tournament
FE-02: View registration status

1.6.2 Limitations & Exclusions

LI-1: Write limitation here.

II. Software Project Management Plan

This chapter describes project planning, scheduling, risk management, communication and configuration management.

2.1. Overview

The Software Project Management Plan (SPMP) outlines the managerial, technical, and resource allocation strategies for the development of the Horse Racing Tournament Management System. This project aims to digitize and automate the processes of managing horse racing tournaments, including horse owner registration, jockey invitation workflows, race scheduling, referee assignment, result recording, and spectator interactions.

The primary objective of this plan is to ensure that the project is delivered on time, within scope, and meets all defined quality criteria. To achieve these goals, the team employs an Agile-Scrum methodology, which promotes iterative development, continuous feedback, and rapid response to changes. This document details the project scope, estimation, quality objectives, risk management, and configuration management processes that govern the lifecycle of this project.

2.2. Scope & Estimation

To ensure the project is delivered successfully within the constraints of time and resources, the project scope has been broken down into a detailed Work Breakdown Structure (WBS). The WBS divides the Horse Racing Tournament Management System into 7 distinct modules, covering administrative tasks, core infrastructure, and functional components for various user roles.

Each work item is assessed for complexity (Simple, Medium, or Complex) based on the database interactions, business logic density, and interface requirements:

- **Simple**: Basic CRUD operations, simple UI layouts, or minor configurations requiring 5 to 6 man-days.
- **Medium**: Standard functional features with moderate business logic (e.g., forms validation, standard data list with filters) requiring 7 to 8 man-days.
- **Complex**: Critical components involving advanced logic, multiple integrations, or strict constraints (e.g., conflict checking algorithms, two-step referee verification, real-time leaderboard calculations) requiring 9 to 15 man-days.

The table below outlines the estimated effort in man-days for each WBS item based on these parameters:

No.	WBS Item	Complexity	Est. Effort (man-days)
1	**PM01. Project Management & Administration**		**20**
1.1	Project Setup & Git Workflow Config	Simple	5
1.2	Code Review, Integration & Conflict Support	Complex	15
2	**DB01. Database & Infrastructure**		**12**
2.1	Database Schema Setup & ERD Design	Medium	7
2.2	API Core Config & DB Connection	Simple	5
3	**FE01. Authentication & User Management**		**14**
3.1	Login, Registration & Authorization UI/API	Medium	8
3.2	User Management Panel (Pagination/Search)	Medium	6
4	**FE02. Horse & Jockey Management**		**23**
4.1	Horse Owner Profile Update & Registration	Medium	8
4.2	Jockey Profile Update & Invitation UI/API	Complex	15
5	**FE03. Tournament & Race Scheduling**		**23**
5.1	Tournament Management & Race Creation	Medium	8
5.2	Race Scheduling & Conflict Validation	Complex	15
6	**FE04. Race Execution & Results**		**24**
6.1	Referee Two-step Result Submission	Complex	12
6.2	Real-time Leaderboard & Prize Config	Complex	12
7	**FE05. Spectator Portal**		**14**
7.1	Schedules & Live Results UI/API	Simple	6
7.2	Place Prediction Form & Timer Lock	Complex	8

Table 2.1: Work Breakdown Structure (WBS) and Effort Estimation

2.3. Project Objectives

To ensure the Horse Racing Tournament Management System meets the quality standards required for deployment and user satisfaction, the team has established specific quality objectives. These objectives focus on functional correctness, performance, security, data integrity, and usability, matching the real development and testing guidelines of the project.

The table below outlines the core quality objectives and target criteria for the project:

No.	Quality Objective	Target Metric / Criteria	Measurement Method
1	Functional Suitability	100% of defined SRS requirements pass manual E2E test cases	Execution of Manual E2E Test Cases (Phase 1, 2, 3, 4)
2	Data Integrity & Encoding	100% successful storage and display of Vietnamese characters (Unicode) in DB and UI	Database inspection (SQL Server) & UI display validation
3	Security & Authentication	Secure password hashing (bcrypt) and JWT authorization for protected role-based routes	Manual code review of app security handlers and route guards
4	Performance Efficiency	Response time < 2s for all frontend page transitions and backend API calls	Network monitoring via Browser Developer Tools
5	Usability & Aesthetics	Responsive dark-mode glassmorphic interface with consistent date/time formats	Manual peer testing & UI checklist validation

Table 2.2: Project Quality Objectives

2.3.1 Defect Tolerance by Testing Phase

Defect density and resolution rates are tracked continuously across various testing levels: Document Review, Integration Testing (Phase 1-2), Integration Testing (Phase 3), and System Testing/UAT. Defects are classified into three severity levels:

- **Critical**: System crash, database connection failure, or core functions completely blocked (e.g., authentication failure, scheduling conflicts, failed result recalculation).
- **Major**: Features malfunctioning or displaying incorrect data without completely blocking the flow (e.g., jockey showing ID instead of real name, date formats displaying raw ISO string).
- **Minor**: UI layout alignment bugs, minor text formatting glitches, or spelling errors that do not affect functional operations.

The table below details the maximum allowed defects and the target resolution process for each testing phase:

Testing Phase	Max Critical	Max Major	Max Minor	Resolution Process
Document Review	0	Max 2	Max 5	Resolved before coding phase
Unit Testing	0	0	Max 3	Resolved before PR merge
Integration Test	0	Max 1	Max 8	Resolved before System Test
System Test	0	Max 2	Max 12	Resolved before UAT phase
Acceptance Test (UAT)	0	0	Max 3	Resolved before final release

Table 2.3: Maximum Allowed Defect Rates by Testing Phase

2.4. Risk Management

To ensure the smooth execution of the Horse Racing Tournament Management System development, the team has identified potential technical, operational, and managerial risks. Managing these risks involves analyzing their probability of occurrence, potential impact, and establishing proactive mitigation and contingency plans.

The table below presents the key risks identified during the project lifecycle and the corresponding response strategies:

No.	Risk Description	Probability	Impact	Mitigation / Response Plan
1	**Git Merge Conflicts**: Code integration conflicts due to multiple developers working on adjacent modules (e.g., dashboard panels).	High	Major	Split large files (like `dashboard/page.js`) into smaller components. Enforce strict branching guidelines: pull changes from `dev-GiaHuy` before coding, and create pull requests for review.
2	**Vietnamese Encoding Errors**: Vietnamese names and text showing as `?` in UI or database due to non-Unicode configurations in SQL Server.	Medium	Moderate	Change string definitions to `Unicode` and text to `UnicodeText` in SQLAlchemy models (`database_models.py`) to map to `NVARCHAR` columns in SQL Server.
3	**Race Scheduling Conflicts**: Double-booking of referees, jockeys, or horses in overlapping races or tournaments.	Medium	Major	Implement robust backend validations (`Conflict Validation`) for schedule updates and referee assignments before persisting to the database.
4	**React Hooks Rule Violations**: Frontend crashes and rendering bugs caused by calling Hooks inside loops or conditional statements (e.g., in Admin panels).	Medium	Major	Perform mandatory code reviews and utilize ESLint static analysis to identify Hook violations prior to code integration.
5	**Inconsistent Local Databases**: Out-of-sync local database schemas among team members causing API integration failures.	High	Major	Provide a centralized database setup and seeding script (`db_setup.py`). Enforce running the script immediately after git pulls.

Table 2.4: Project Risk Analysis and Mitigation Plans

2.5. Management Approach

To effectively manage the development of the Horse Racing Tournament Management System, the team employs the Agile-Scrum methodology. This approach allows the team to deliver features iteratively, adapt to changes quickly, and ensure constant communication among members.

2.5.1 Agile-Scrum Implementation

The development cycle is organized into weekly sprints. Each sprint begins with a planning meeting and ends with a review and retrospective.

- **Sprint Planning**: Held at the start of each week to define the sprint goal and allocate tasks from the product backlog on Jira.
- **Weekly Syncs**: Held twice a week (on Wednesdays and Saturdays) to track progress, identify blockers, and align integration efforts. Daily updates and progress reports are shared via Zalo.
- **Sprint Review & Retrospective**: Conducted at the end of the sprint to review completed tasks, demo features, and reflect on improvements for the next sprint.

2.5.2 Coding Standards and Bug Prevention

To maintain code quality and prevent bugs early in the lifecycle, the team enforces the following rules and standards:

- **Coding Style Guidelines**: The backend follows PEP 8 guidelines for Python (FastAPI), while the frontend adheres to ESLint standards for React/Next.js. React Hooks rules are strictly monitored (e.g., avoiding calling hooks inside loops or conditions).
- **Database Consistency**: Any database schema modifications must be updated in the SQLAlchemy models. Developers are required to run the automated database setup and seeding script (`db_setup.py`) locally after pulling the latest code to prevent database mismatches.
- **Git Workflow Rules**: Direct pushes to the main branch (`main`) or integration branch (`dev-GiaHuy`) are prohibited. All work must be completed on feature branches (`feature/*`) and integrated via Pull Requests (PRs). The Team Leader reviews and approves all PRs before merging.

2.6. Deliverables Schedule

The project deliverables are scheduled and tracked weekly, aligning with the Agile sprints and the development phases of the Horse Racing Tournament Management System. The following schedule details the deliverables for both source code components (API and UI) and project documentation from Week 1 to Week 9.

2.6.1 Deliverables Timeline (Week 1 - Week 9)

The timeline below structures the actual development history, pull requests, and documentation milestones into weekly progress updates from the beginning of the project through the final LaTeX compilation:

Week	Phase / Focus	Key Deliverables (Source Code & Documentation)	Status
Week 1	Scope & Context	System Scope definition, Actors identification, Context Diagram, and initial database setup (`schema.sql`).	Completed
Week 2	Requirements	Overall Use Case Diagram for MVP, Use Case Descriptions, and initial project layout setup.	Completed
Week 3	Phase 1 (Core Auth)	Next.js and FastAPI project skeleton, user registration, login flow, and dashboard layout restructuring.	Completed
Week 4	Phase 1 (Admin Flow)	User Management API, lock/unlock accounts, dynamically calling referees list, and fixing React Hooks rendering violations. Merged PR #4 and #6 (16/06/2026).	Completed
Week 5	Phase 2 (Owner & Jockey)	Jockey invitation workflow (Accept/Reject), horse profiles CRUD with age validation (2-10 years), and registration status updates. Merged (19/06/2026).	Completed
Week 6	Phase 3 (Referee & Spectator)	Two-step result verification (`RESULTS_ENTERED` → `COMPLETED`), `RaceInspections` form, spectator place prediction form with dynamic filters, and auto-reward points logic. Merged (21/06/2026).	Completed
Week 7	Phase 4 (Admin & Database)	Timezone `Asia/Ho_Chi_Minh` integration, ±2h referee conflict check, validation rules, database schema update (missing columns for users, spectators), and fixing local connections (127.0.0.1). Merged (24/06/2026).	Completed
Week 8	LaTeX	LaTeX compilation environment setup, project structure creation, and initial drafts of Chapter 1 and 2 sections.	Completed
Week 9	LaTeX Compilation	Finalizing Chapter 2 (SPMP) sections, Chapter 3 (SRS) requirements, Chapter 4 (SDD) system design, and compiling the official PDF document.	Completed

Table 2.5: Key Deliverables Schedule (Week 1 - 9)

2.7. Responsibility Assignment Matrix (DRSI)

To clarify roles and responsibilities across the team, the project utilizes a DRSI matrix (Do, Review, Support, Informed). The abbreviations used in the matrix are defined as follows:

- **D (Do)**: The member responsible for developing and implementing the deliverable.

- **R (Review)**: The member responsible for reviewing and approving the quality of the work.
- **S (Support)**: The member providing support during development or integration.
- **I (Informed)**: Members kept updated on the status of the deliverable.
- **- (Omitted)**: Member has no direct responsibility for the deliverable.

The table below outlines the DRSI matrix for the key project deliverables:

Deliverable / Module	DH	TA	GH	TC	MH	BH	TM
Project Setup & Configuration	D	I	I	I	I	I	I
Database & Infrastructure Setup	R	I	D	I	I	S	I
Authentication & User Management	R	I	D	I	D	I	I
Horse Owner Registration	R	D	I	I	I	I	I
Jockey Profile & Invitations	R	S	I	D	I	I	I
Race Scheduling & Referee Assign	R	I	S	I	D	I	I
Race Result & Ranking Calculation	R	I	I	I	I	D	I
Spectator Prediction & Rewards	R	I	I	I	I	S	D
LaTeX Report & Presentation	D	D	D	D	D	D	D

Table 2.6: DRSI Responsibility Assignment Matrix

2.7.1 Document Management

To ensure consistent document formatting and prevent conflicts during collaborative writing, the team follows a structured LaTeX document management workflow on GitHub:

- **Three-Tier Architecture**: The document structure is divided into three tiers:
  1. `main.tex`: The root file that only handles package imports and includes document wrappers.
  2. `documents/`: Wrapper files for each chapter (e.g., `project_management_plan.tex`).
  3. `sections/`: The folder containing the actual content files divided by section and assignee.
- **Conflict Avoidance**: Team members are strictly restricted to editing files under their assigned sections. Editing `main.tex` or files in the `documents/` folder is prohibited without prior team alignment.
- **Figure Convention**: Figures must be stored in the `figures/` directory and named according to the format: `<chapter>_<content>.<ext>` (e.g., `figures/srs_erd.png`). Generic names like `image1.png` are avoided.

2.7.2 Source Code Management

The project source code is hosted on GitHub and managed using a customized feature-branching workflow to guarantee code quality and repository stability:

- **Branching Strategy**:
  - `main`: Reserved for the final, production-ready release used for demos.
  - `dev-GiaHuy`: The main integration branch where all validated feature branches are merged.
  - `feature/*`: Temporary branches used by developers to work on assigned tasks (e.g., `feature/auth-user`, `feature/spectator-view`).
- **Development Protocol**:
  1. Pull the latest updates from the remote integration branch before coding: `git pull origin dev-GiaHuy`.
  2. Develop the feature on a separate `feature/*` branch.
  3. Push the feature branch to the remote repository.
  4. Open a Pull Request (PR) from the feature branch to `dev-GiaHuy`. Direct pushes to `dev-GiaHuy` or `main` are disabled.
- **Commit Message Convention**: Commit messages must clearly state the scope of the change. For example:
  - `git commit -m "Add login API"`
  - `git commit -m "Create horse management page"`
- **Review & Merge**: The Team Leader reviews all PRs, checks for build errors, resolves conflicts, and merges the code into the integration branch.

2.7.3 Tools and Infrastructure

The development and execution environment for the Horse Racing Tournament Management System utilizes the following tech stack, tools, and local infrastructure:

- **Programming Languages & Frameworks**:
  - **Backend**: Python 3 utilizing the FastAPI framework for clean architecture REST APIs.
  - **Frontend**: JavaScript/React.js utilizing Next.js (App Router) for the web application UI.
- **Database Management**:
  - Local Microsoft SQL Server Express instance (`localhost\SQLEXPRESS`) with database named `HorseRacing`.
  - Connection library: `pyodbc` (ODBC Driver 17 for SQL Server) and SQLAlchemy ORM.

- **Development Platforms**:
  - **Operating System**: Windows 10/11 workstation.
  - **Integrated Development Environment (IDE)**: VS Code equipped with LaTeX Workshop extension.
  - **Git Client**: Git for Windows / Git Bash.
  - **LaTeX Compiler**: MiKTeX or TeX Live for generating report PDFs locally.
- **Local Hosting Configuration**:
  - Backend server running locally on `http://127.0.0.1:8000` (API Client configured directly to IP address to prevent connections dropouts).
  - Frontend Next.js development server running locally on `http://localhost:3000`.

III. Software Requirement Specification

3.1. Product Overview

The Horse Racing Tournament Management System (HRTMS) is a comprehensive web-based application designed to digitize and optimize the operational workflows of horse racing tournament organizations, race scheduling, and spectator participation.

- **Operational Environment**:
  - **Platform**: Web-based application accessible via standard web browsers (Chrome, Edge, Firefox, Safari).
  - **Connectivity**: Requires a stable Internet connection to function (no offline mode).
  - **Infrastructure**: Powered by a FastAPI backend API and a Next.js frontend, utilizing Microsoft SQL Server for the database.
- **Anticipated Users**:
  - **Internal Users**: Administrator and Race Referees.
  - **External Users**: Horse Owners, Jockeys, and Spectators.
- **Constraints & Assumptions**:
  - **Language**: The system database and interface support UTF-8 encoding for Vietnamese characters, with localized date/time formatting.
  - **Compliance**: Complies with user account data privacy and local tournament scheduling rules in Vietnam.
  - **Scope Limitation**: Does not support actual financial betting transactions; spectator predictions use virtual reward points only. Timezone handling is restricted to `Asia/Ho_Chi_Minh`.
- **Dependencies**:
  - Relies on local Python environment libraries (`pyodbc`, `SQLAlchemy`, `FastAPI`).
  - Requires a configured Microsoft SQL Server database.

3.1.1 System Context Diagram

The system context diagram below defines the boundaries and exchanges between the Horse Racing Tournament Management System and its external entities:

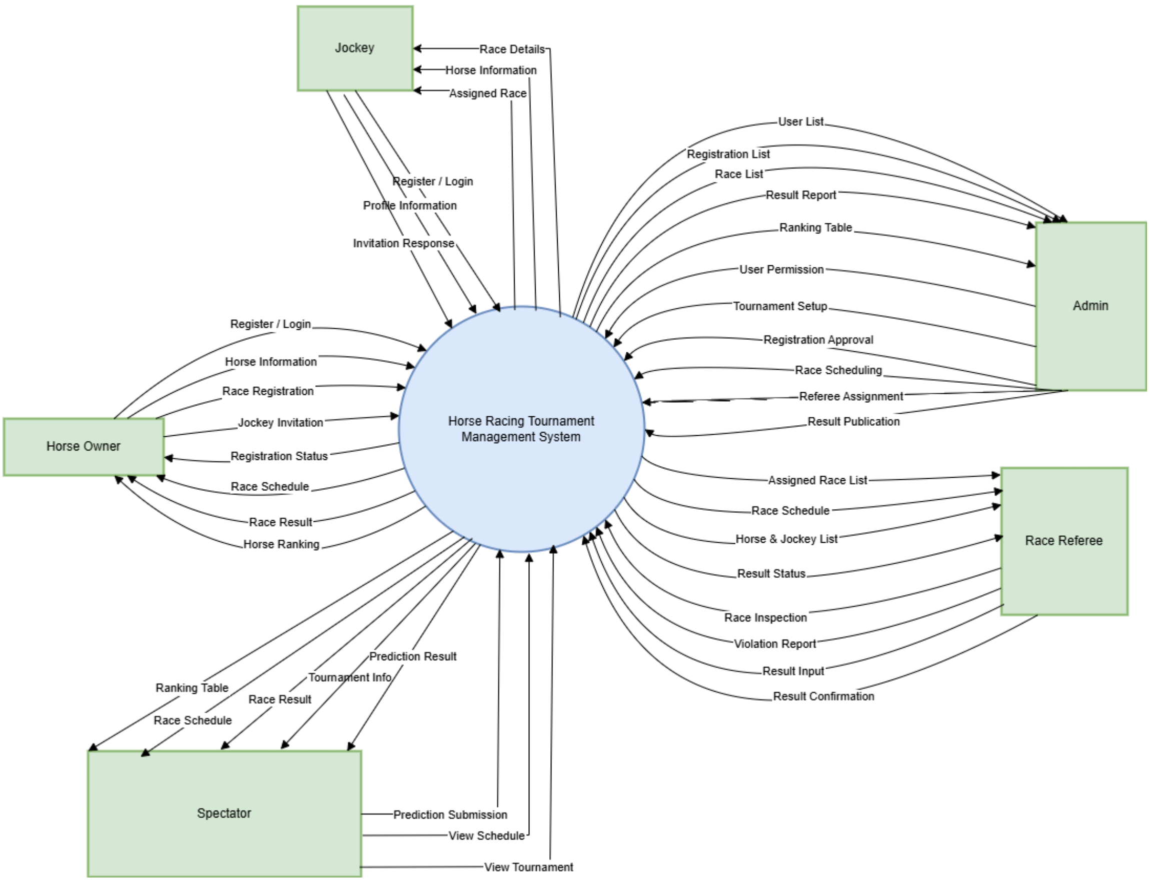


Figure 3.1: System Context Diagram

3.1.2 Interaction Data Flows

The data interactions between the system and each external actor are defined as follows:

- **Horse Owner**:
  - *Data Input to System*: Register / Login, Horse Information, Race Registration, and Jockey Invitation.
  - *Data Output from System*: Registration Status, Race Schedule, Race Result, and Horse Ranking.
- **Jockey**:
  - *Data Input to System*: Register / Login, Profile Information, and Invitation Response.
  - *Data Output from System*: Assigned Race, Race Details, and Horse Information.
- **Administrator (Admin)**:

- *Data Input to System*: User Permission, Tournament Setup, Registration Approval, Race Scheduling, Referee Assignment, and Result Publication.
- *Data Output from System*: User List, Registration List, Race List, Result Report, and Ranking Table.

- **Race Referee**:
  - *Data Input to System*: Race Inspection, Violation Report, Result Input, and Result Confirmation.
  - *Data Output from System*: Assigned Race List, Race Schedule, Horse & Jockey List, and Result Status.

- **Spectator**:
  - *Data Input to System*: View Tournament, Prediction Submission, and View Schedule.
  - *Data Output from System*: Tournament Info, Race Schedule, Race Result, Ranking Table, and Prediction Result.

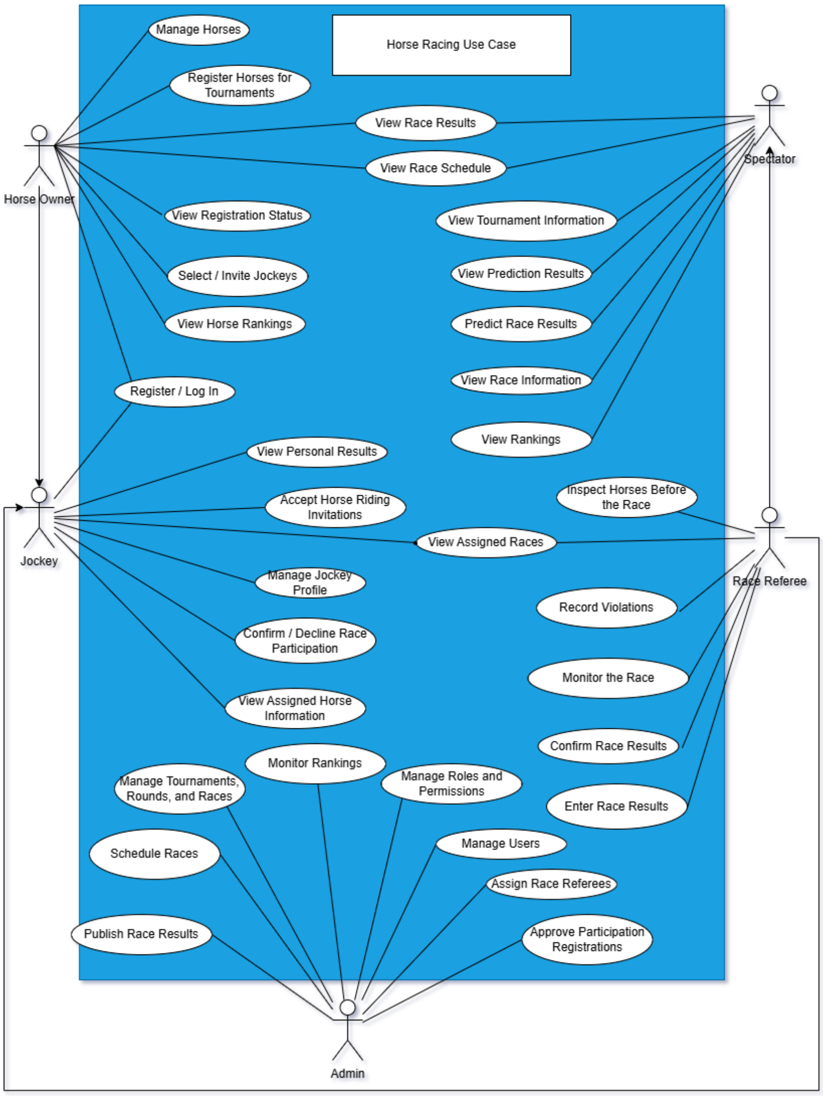
3.2. User Requirements

3.2.1 System Actors

Actor	Type	Description / Responsibilities
Admin	Human	Responsible for platform administration: user and role management, system configuration, approvals, and overall system monitoring.
Referee	Human	Manages race supervision and validation: verifies race results, resolves disputes, and updates official results (backend endpoints in ‘results.py‘ / ‘referees.py‘).
Jockey	Human	Represents the rider; manages jockey profile information and race assignments.
Horse Owner	Human	Registers and manages horse profiles, submits horses to races, and maintains horse-related records and health information.
Spectator	Human	Views race schedules, results, leaderboards, and (optionally) places bets or follows favorite horses/jockeys via the frontend panels.

3.2.2 Use Cases

Use Case Diagram



Use Case Description

ID	Use Case	Primary Actor	Brief Description
UC-01	Manage Horses	Horse Owner	Create, update and maintain horse profiles and related health records.
UC-02	Register Horses for Tournaments	Horse Owner	Submit horses to tournament participation and view registration status.
UC-03	Manage Jockey Profile	Jockey	Create and update jockey personal/profile information and availability.
UC-04	Confirm / Decline Race Participation	Jockey	Accept or decline invitations to participate in assigned races.
UC-05	View Race Schedule	Spectator	Browse upcoming races, schedules and basic race information.
UC-06	View Race Results	Spectator	View final results, rankings and statistics after a race is published.
UC-07	Predict / View Prediction Results	Spectator	Submit predictions (if supported) and view prediction outcomes.
UC-08	Inspect Horses / Monitor Race	Referee	Inspect horses before the race, monitor the race and record any violations.
UC-09	Enter / Confirm Race Results	Referee	Enter raw race results and confirm official results for publication.
UC-10	Manage Races and Tournaments	Admin	Create and schedule races/tournaments, assign referees and publish results.
UC-11	Manage Users and Roles	Admin	Manage platform users, roles and permissions (admin panel).

3.3. Functional Requirements

3.3.1 System Functional Overview

Admin Screen Flow

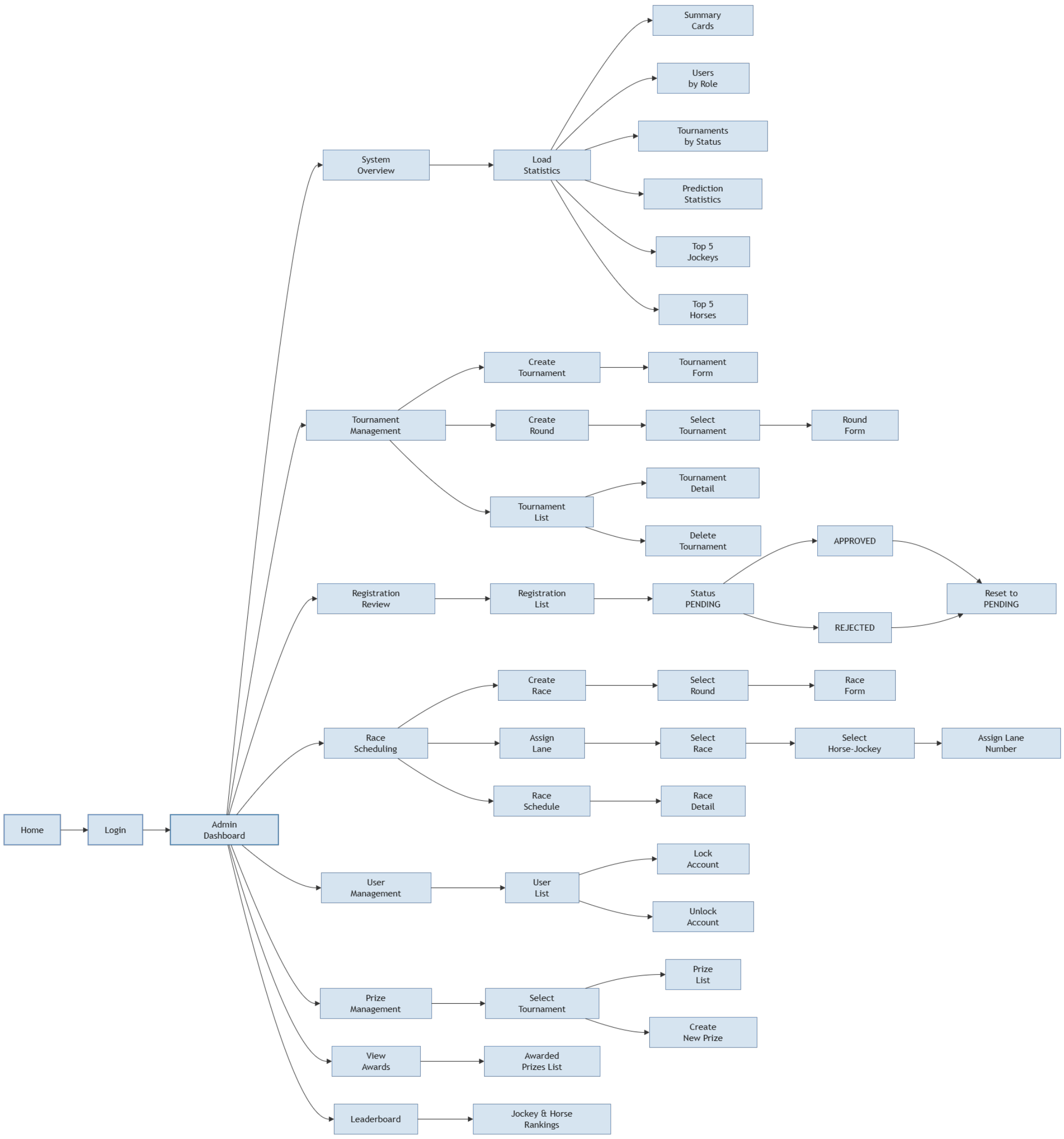


Figure 3.2: Admin screen flow

Admin Screen description

#	Feature	Screen	Description
1	Navigation	Home	Initial landing page that leads to the login screen for an admin user.
2	Authentication	Login	Secure admin sign-in screen with credentials and access validation.
3	Dashboard	Admin Dashboard	Main admin hub that provides links to system overview, management panels, and key actions.
4	Overview	System Overview	Displays the high-level system summary and provides access to statistics modules.
5	Statistics	Load Statistics	Shows dashboard widgets such as summary cards, user counts by role, tournament status, prediction metrics, top jockeys, and top horses.
6	Tournament Management	Tournament List	Lists tournaments with controls to view details, approve or reject, and delete tournaments.
7	Tournament Management	Tournament Form	Screen for creating a new tournament with required details and rules.
8	Tournament Management	Round Form	Screen for creating a new tournament round after selecting a tournament.
9	Registration Review	Registration List	Displays pending horse registration requests and allows the admin to review status.
10	Registration Review	Registration Status	Shows the current approval state of a registration, including options to approve, reject, or reset to pending.
11	Race Scheduling	Create Race	Screen for creating a new race based on selected tournament rounds.
12	Race Scheduling	Assign Lane	Allows the admin to assign a lane number to a selected horse-jockey pair for a specific race.
13	Race Scheduling	Race Schedule	Displays the full race schedule and navigation to race detail pages.
14	User Management	User List	Lists platform users with options to lock or unlock accounts and manage roles.
15	Prize Management	Prize List	Shows existing prize entries for a selected tournament and management actions.
16	Prize Management	Create New Prize	Screen to add a new prize associated with a selected tournament.
17	Rankings	Leaderboard	Displays jockey and horse rankings and performance standings.

Table 3.1: Admin screen descriptions

Horse Owner Screen Flow

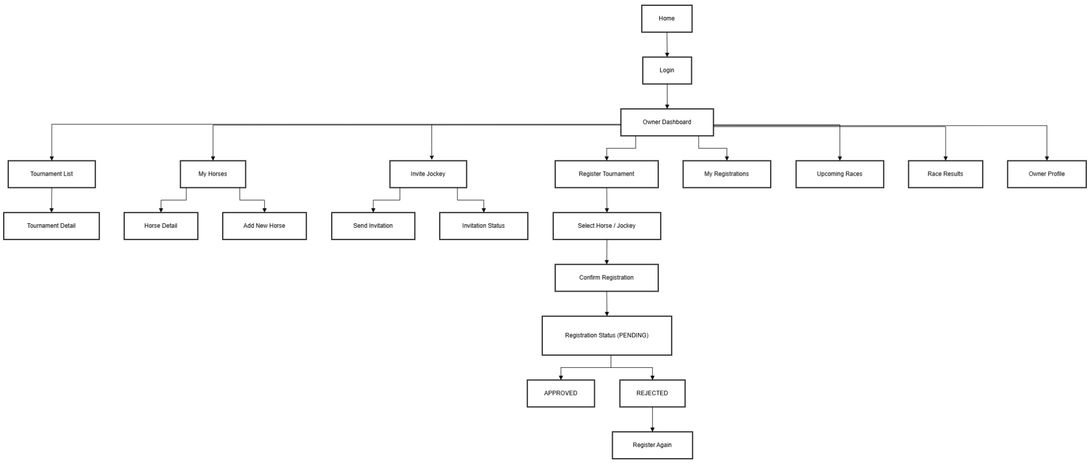


Figure 3.3: Horse Owner screen flow

Horse Owner Screen description

#	Feature	Screen	Description
1	General	Home	Landing page for all users; navigate to Login and Owner access.
2	General	Login	Authenticate the owner and enter the Owner Dashboard.
3	Dashboard	Owner Dashboard	Main hub for Owner with navigation to horse management, invitations, tournament registration, schedules, results, and profile.
4	Horse Management	My Horses	View and manage owned horses, including add, edit, and delete operations.
5	Horse Management	Horse Detail	Edit horse details: name, age (2-10), breed, gender, and save changes.
6	Invitation	Invite Jockey	Send invitations to jockeys for a selected horse and tournament with a custom message.
7	Invitation	Invitation Status	Track invitations with jockey, horse, tournament, and current acceptance status.
8	Registration	Register Tournament	Register accepted horse/jockey pairs for tournaments and create registration requests.
9	Registration	My Registrations	Display tournament registration history with approval status: APPROVED, PENDING, REJECTED.
10	Schedule	Upcoming Races	Show upcoming races for the owner's horses with date, time, and location.
11	Results	Race Results	Show historical race results for the owner's horses with rank, points, notes, and violations.
12	Profile	Owner Profile	View and update owner details: full name, phone, company, avatar.

Table 3.2: Horse Owner screen descriptions

Jockey Screen Flow

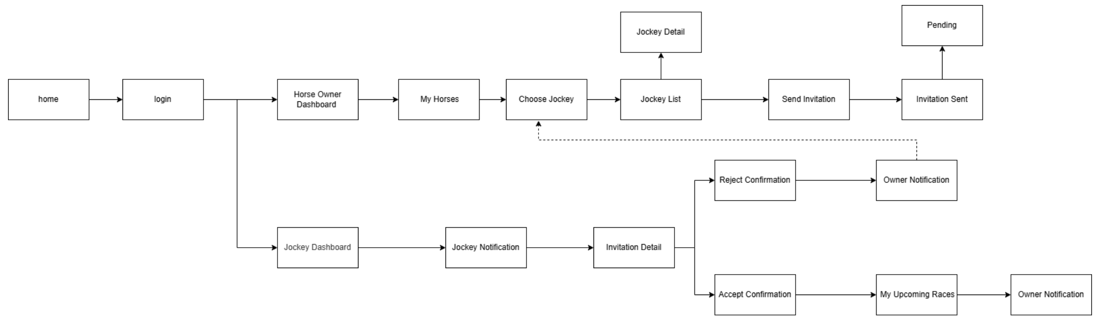


Figure 3.4: Jockey screen flow

Jockey Screen description

#	Feature	Screen	Description
1	General	Home	Landing page for all users; navigate to Login.
2	General	Login	Authenticate the jockey and enter the Jockey Dashboard.
3	Dashboard	Jockey Dashboard	Main hub for jockey with navigation to invitations, upcoming races, results, and profile.
4	Notifications	Jockey Notification	View pending invitations from owners with horse, owner, tournament, and current status.
5	Invitation	Invitation Detail	Display invitation details including owner, horse, tournament, schedule, and custom message.
6	Invitation	Accept Confirmation	Confirm acceptance of the invitation and register the jockey for the selected race.
7	Invitation	Reject Confirmation	Decline the invitation and send notification back to the owner.
8	Schedule	My Upcoming Races	Show accepted races with date, time, horse, owner, tournament, and race status.
9	Notification	Owner Notification	Notify the owner whether the invitation was accepted or rejected.

Table 3.3: Jockey screen descriptions

Race Scheduling Management Screen Flow

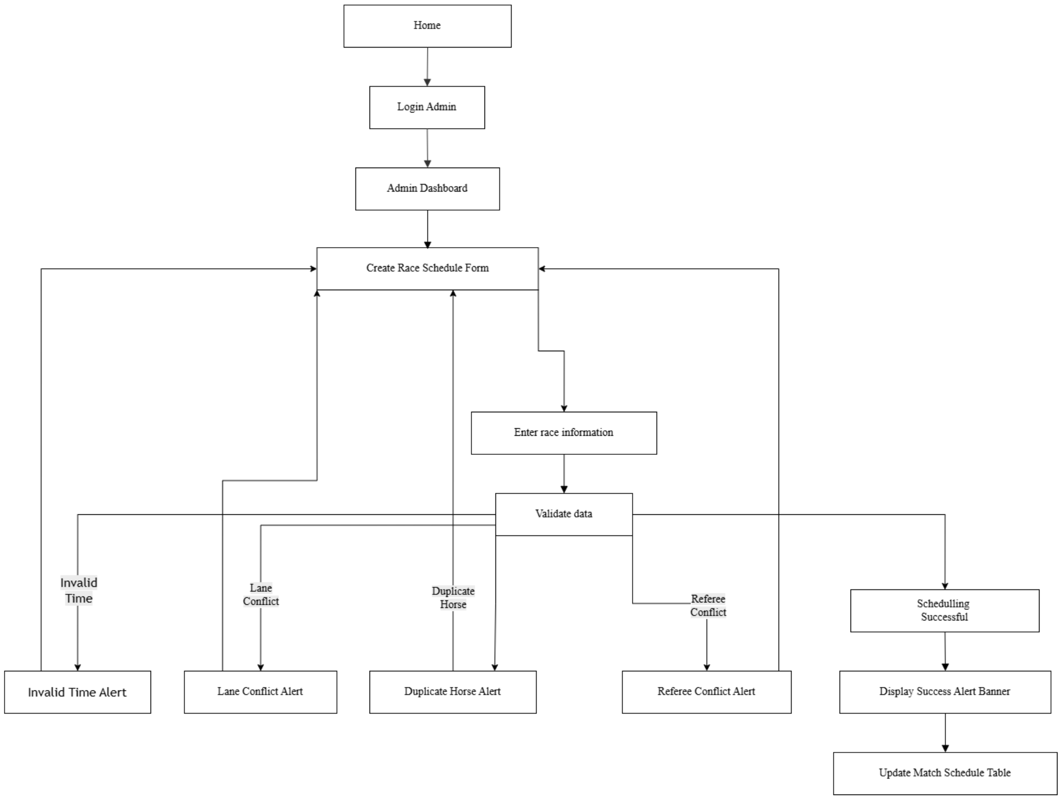


Figure 3.5: Race scheduling management screen flow

Screen description for Race Scheduling

#	Feature	Screen	Description
1	General	Home	The main landing page of the system, acting as the entry point before authentication.
2	Authentication	Login Admin	The dedicated login interface restricted to authorized Administrator accounts.
3	General	Admin Dashboard	The landing page post-login; displays the ongoing tournament list to select a specific workspace.
4	Configuration	Create Race Schedule Form	Main input interface to set up tournament stage, race name, time, track conditions, referee assignment, and lane allocation.
5	Configuration	Enter race information	Step where detailed race attributes are inputted before the system initiates the validation process.
6	Validation	Validate data	Automated backend process triggered to verify the validity of data and check for any constraints or scheduling conflicts.
7	Validation	Invalid Time Alert	Validation error state triggered if the selected race time falls outside the boundaries of the tournament start and end dates.
8	Validation	Lane Conflict Alert	Validation error state triggered if multiple horses are mistakenly assigned to the exact same lane number.
9	Validation	Duplicate Horse Alert	Validation error state triggered if a single horse is selected more than once within the same race setup.
10	Validation	Referee Conflict Alert	Validation error state triggered if the assigned referee has another active commitment within a 2-hour window.
11	Validation	Scheduling Successful	System state confirming that the schedule data passed all validations and has been successfully recorded.
12	Validation	Display Success Alert Banner	A success banner prompted at the top of the form displaying: "Race created successfully!".
13	Feedback	Update Match Schedule Table	Persistent historical data table below the form that appends and displays the newly scheduled race details instantly.

Table 3.4: Race Scheduling screen descriptions

Spectator Screen Flow

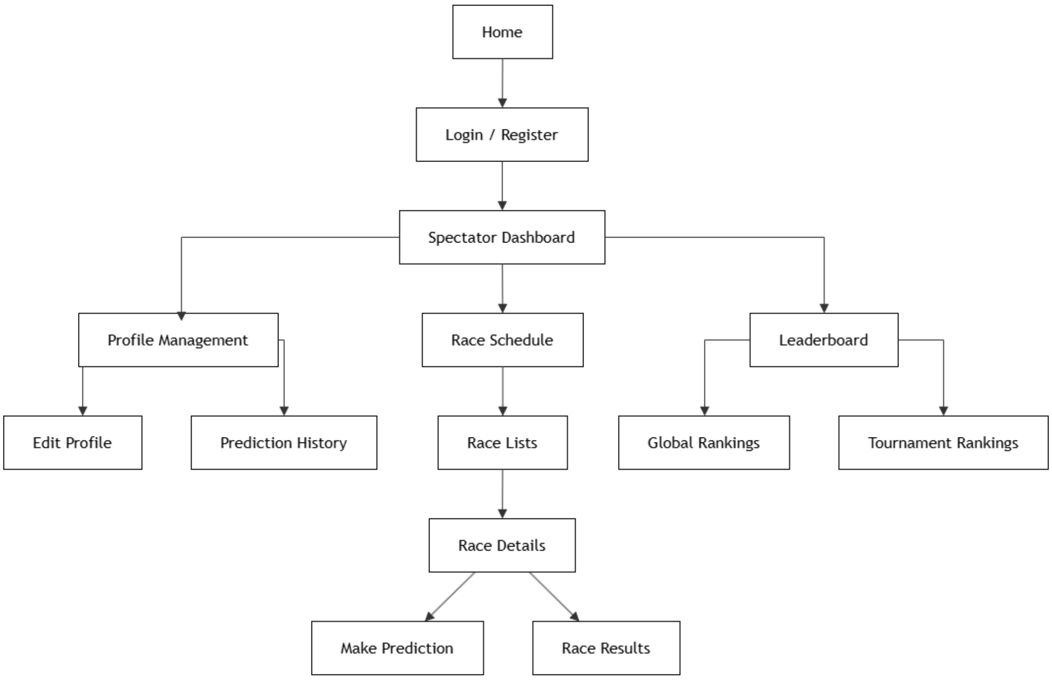


Figure 3.6: Spectator screen flow

Spectator Screen description

#	Feature	Screen	Description
1	General	Home	Landing page for all users; navigate to Login and Registration access.
2	General	Login / Register	Authenticate spectators or register a new account to access the system.
3	Dashboard	Spectator Dashboard	Main hub for Spectator with navigation shortcuts to profile management, race schedules, and leaderboards.
4	Profile	Profile Management	Manage personal account information, providing options to edit profile details and view prediction history.
5	Profile	Edit Profile	Edit and update spectator profile details (full name, phone number, avatar, etc.).
6	Profile	Prediction History	Review the history of placed race predictions along with their corresponding earned points status.
7	Schedule	Race Schedule	View an overview schedule of upcoming tournaments and race rounds.
8	Schedule	Race Lists	Display the detailed list of specific races under the selected schedule.
9	Schedule	Race Details	View comprehensive details of a specific race (participants, time, status) with options to make predictions or view results.
10	Schedule	Make Prediction	Place predictions on results or rankings for upcoming/unstarted races.
11	Schedule	Race Results	View the final outcomes, standings, and technical statistics of a completed race.
12	Leaderboard	Leaderboard	Main section to view rankings, performance, and scores.
13	Leaderboard	Global Rankings	Display the global leaderboard based on total accumulated user points.
14	Leaderboard	Tournament Rankings	Display detailed standings categorized by specific tournaments.

Table 3.5: Spectator screen descriptions

3.3.2 Web Application

Horse Owner Login

Role: Horse Owner

- **Function trigger:** The owner enters account credentials to log in.
- **Function description:**
  - **Purpose:**
    1. Verify the owner's identity.
    2. Grant access to owner functionalities.
  - **Data processing:**
    1. The owner enters username and password.
    2. The system validates credentials.
    3. The system identifies the OWNER role.
    4. The system creates a login session.
- **Function details:**
  - Successful login redirects to Owner Dashboard.
  - Invalid credentials display an error message.
  - Users without accounts may register.

Screen layout:

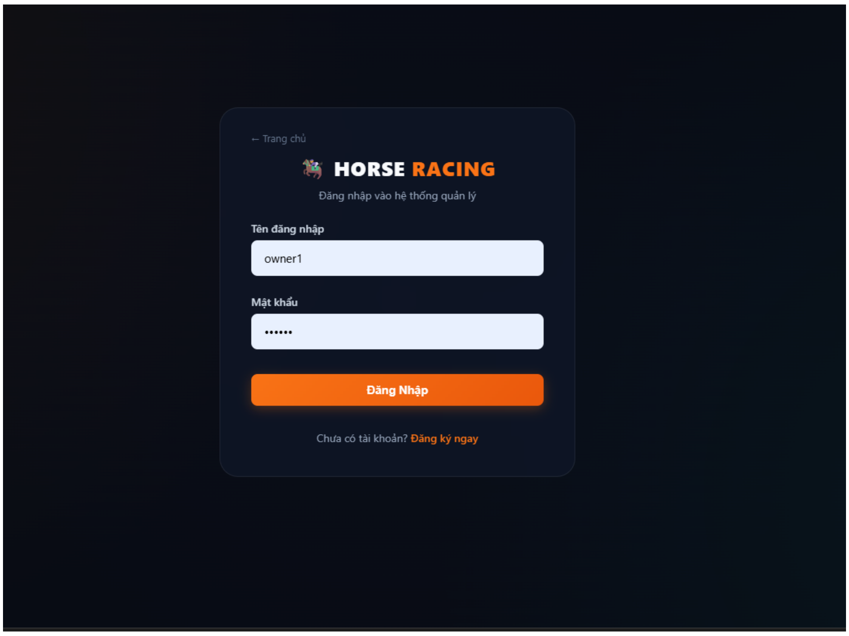


Figure 3.7: Horse Owner Login

Manage Horses

Role: Horse Owner

- **Function trigger:** Select "Quan ly Ngua".
- **Function description:**

- **Purpose:**
  1. Register new horses.
  2. Manage owned horses.
- **Data processing:**
  1. Enter horse name.
  2. Enter age.
  3. Select breed.
  4. Select gender.
  5. Save horse information.

- **Function details:**
  - Add new horses.
  - Edit horse information.
  - Delete horse information.
  - Display horse list.

Screen layout:

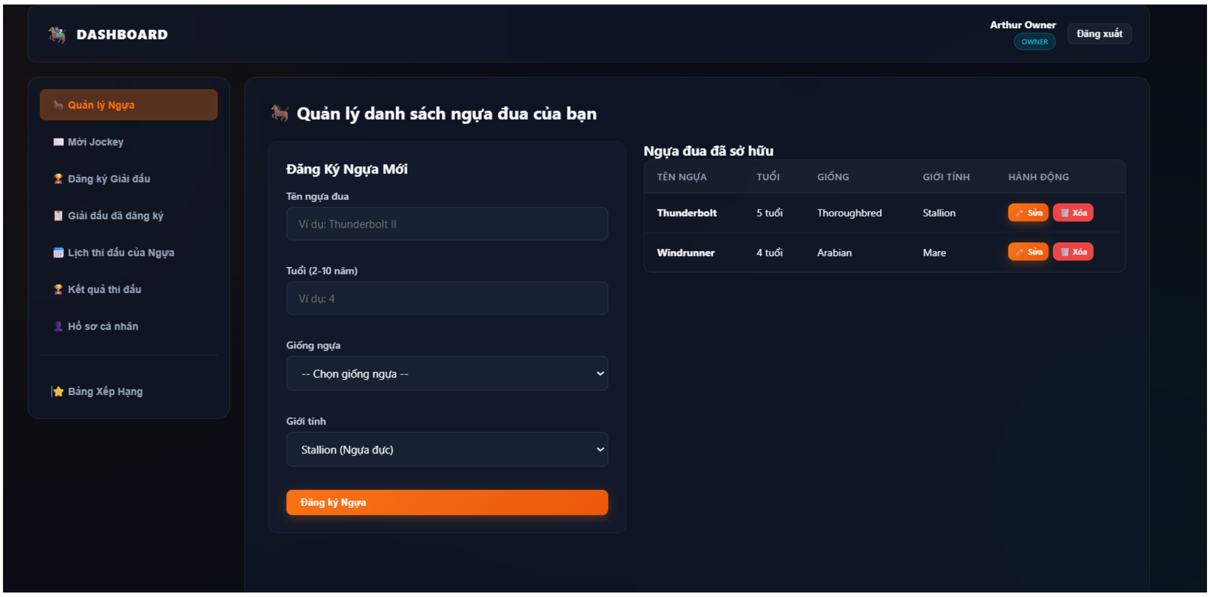


Figure 3.8: Manage Horses

Invite Jockey

Role: Horse Owner

- **Function trigger:** Select “Moi Jockey”.
- **Function description:**
  - **Purpose:**

1. Invite jockeys.
2. Create horse-jockey partnerships.

– **Data processing:**

1. Select jockey.
2. Select horse.
3. Select tournament.
4. Enter invitation message.
5. Send invitation.

- **Function details:**
  – Invitation status is displayed.
  – Pending invitations can be monitored.
  – Accepted invitations may be used for registration.

Screen layout:

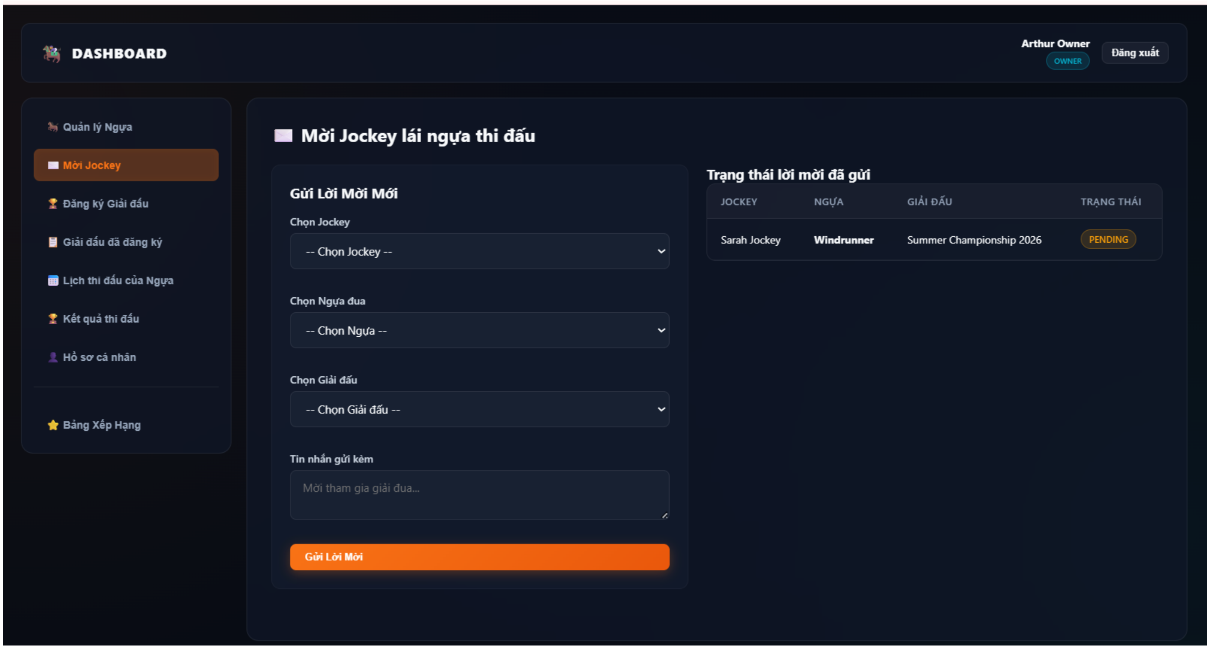


Figure 3.9: Invite Jockey

Register Tournament

Role: Horse Owner

- **Function trigger:** Select “Dang ky Giai dau”.
- **Function description:**
  – **Purpose:**

1. Register horse-jockey pairs.

2. Participate in tournaments.

– **Data processing:**

1. Retrieve available tournaments.
2. Check accepted jockey invitations.
3. Select horse-jockey pair.
4. Submit registration request.

- **Function details:**

  – Only accepted invitations may register.
  – Registration requests are sent to administrators.
  – Invalid pairs cannot participate.

Screen layout:

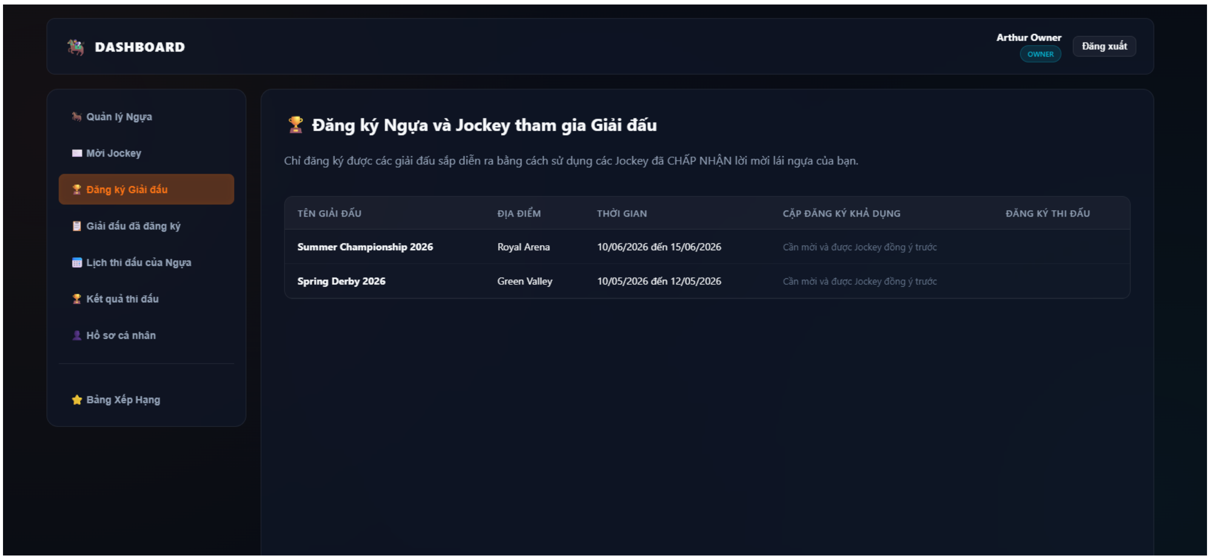


Figure 3.10: Tournament Registration

Registered Tournaments

Role: Horse Owner

- **Function trigger:** Select "Giai dau da dang ky".
- **Function description:**

  – **Purpose:**

  1. Monitor registration status.
  2. View approval results.

  – **Data processing:**

  1. Retrieve registration records.
  2. Retrieve approval status.

3. Display registration history.

- **Function details:**
  - Display horse information.
  - Display jockey information.
  - Display approval status.

Screen layout:

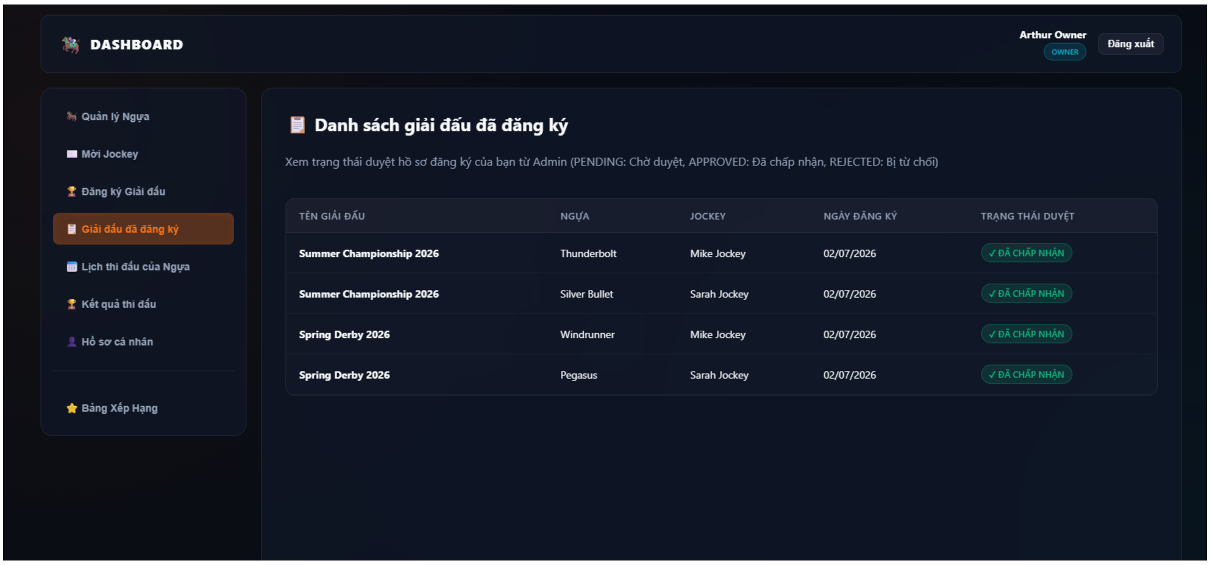


Figure 3.11: Registered Tournaments

Horse Schedule

Role: Horse Owner

- **Function trigger:** Select “Lich thi dau cua Ngua”.
- **Function description:**
  - **Purpose:**
    1. View upcoming races.
    2. Prepare for competitions.
  - **Data processing:**
    1. Retrieve race schedules.
    2. Retrieve tournament information.
    3. Display race information.
- **Function details:**
  - Display race date.
  - Display location.

– Display tournament information.

Screen layout:

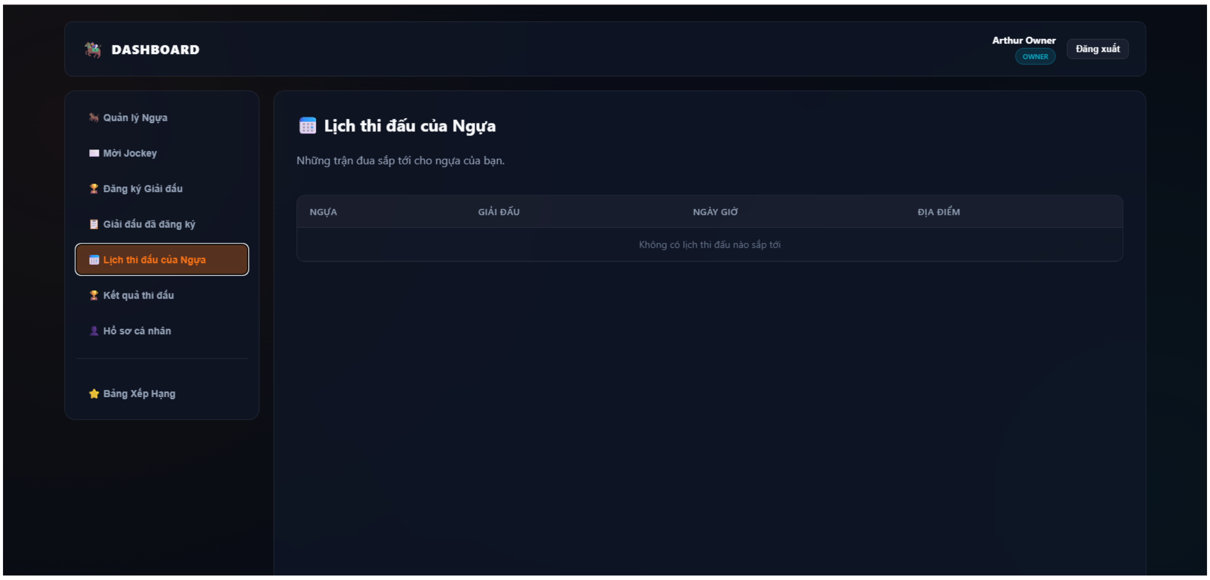


Figure 3.12: Horse Schedule

Race Results

Role: Horse Owner

- **Function trigger:** Select "Ket qua thi dau".
- **Function description:**
  - **Purpose:**
    1. Review race results.
    2. Evaluate horse performance.
  - **Data processing:**
    1. Retrieve completed races.
    2. Retrieve rankings.
    3. Retrieve scores.
    4. Retrieve violations.
- **Function details:**
  - Display race rankings.
  - Display accumulated points.
  - Display remarks.
  - Display violations.

Screen layout:

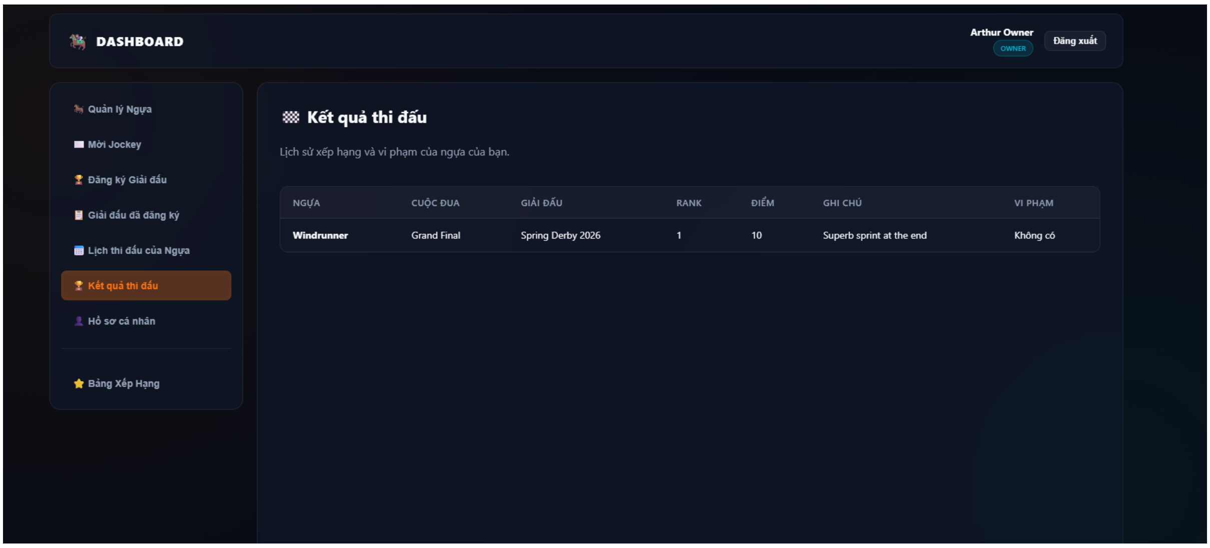


Figure 3.13: Race Results

Owner Profile

Role: Horse Owner

- **Function trigger:** Select "Ho so ca nhan".
- **Function description:**
  - **Purpose:**
    1. Manage personal information.
    2. Update profile information.
  - **Data processing:**
    1. Update personal information.
    2. Update contact information.
    3. Update company information.
    4. Save profile data.
- **Function details:**
  - Edit owner profile.
  - Edit biography.
  - Update social information.

Screen layout:

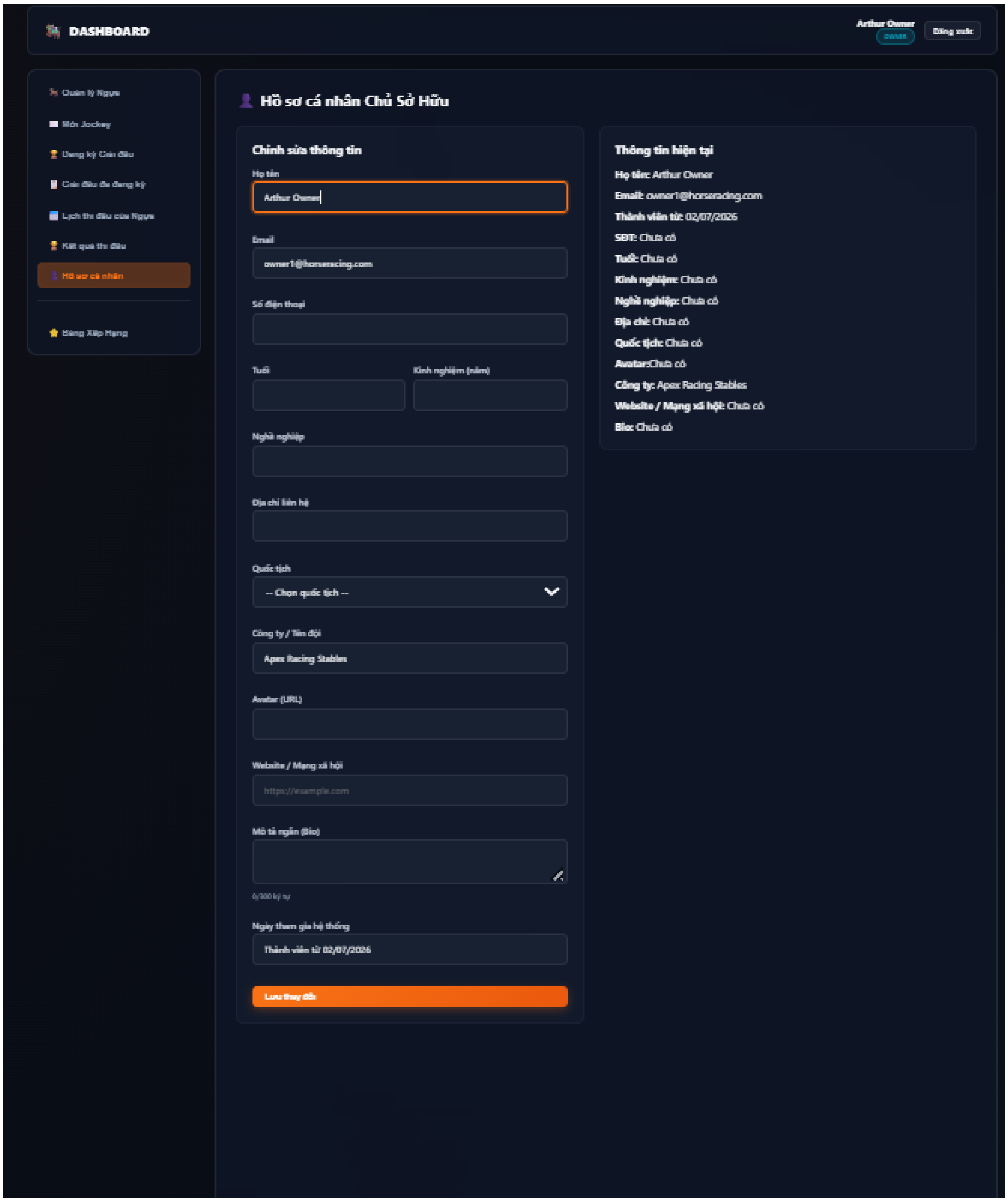


Figure 3.14: Owner Profile

Ranking

Role: Horse Owner

- **Function trigger:** Select "Bang xep hang".
- **Function description:**
  - **Purpose:**
    1. View official rankings.
    2. Compare competition performance.
  - **Data processing:**

1. Retrieve horse rankings.
2. Retrieve jockey rankings.
3. Filter tournaments.

- **Function details:**
  - Display ranking positions.
  - Display accumulated scores.
  - Display tournament filters.

Screen layout:

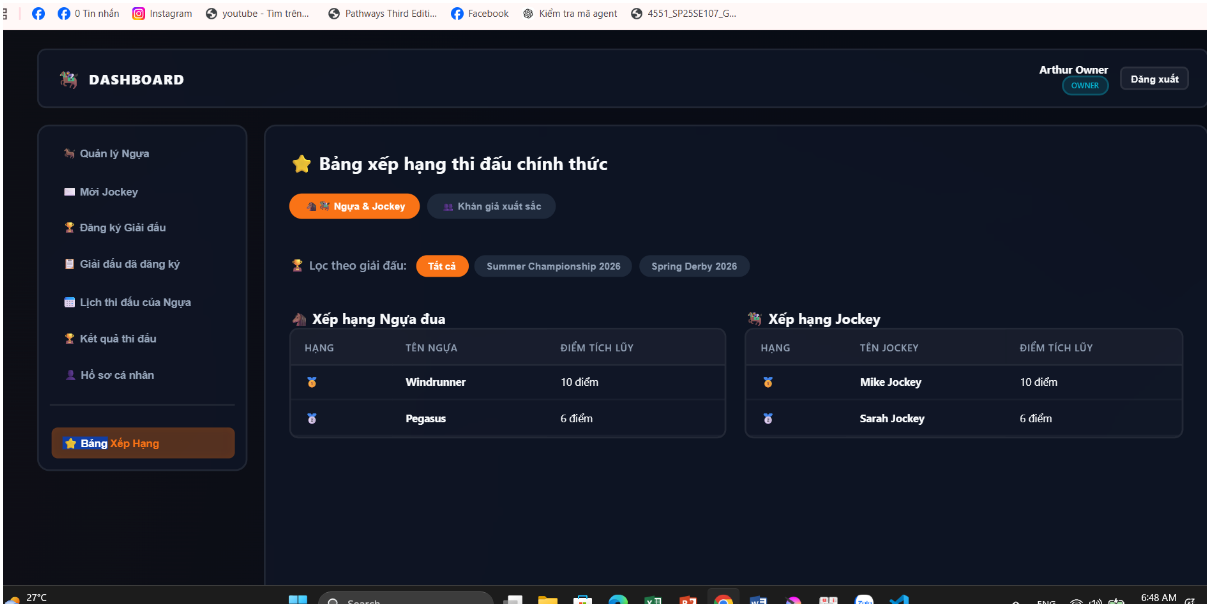


Figure 3.15: Ranking

Spectator Login

Role: Spectator

- **Function trigger:** The spectator enters account credentials to log in.
- **Function description:**
  - **Purpose:**
    1. Verify the spectator's identity.
    2. Grant access to spectator functionalities.
  - **Data processing:**
    1. The spectator enters username and password.
    2. The system validates credentials.
    3. The system identifies the SPECTATOR role.
    4. The system creates a login session.

Screen layout:

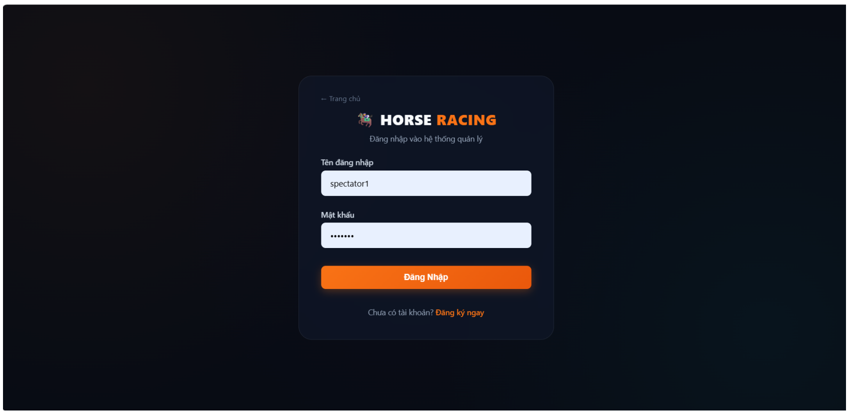


Figure 3.16: Spectator Login

Prediction Management

Role: Spectator

- **Function trigger:** The spectator navigates to the "Du doan Tran dua" (Predict Race) menu.
- **Function description:**
  - **Purpose:**
    1. Allow the spectator to submit a new race ranking prediction to earn reward points.
    2. Allow the spectator to view, edit, or delete their existing prediction history.
  - **Data processing:**
    1. **For creating a new prediction:**
       * The spectator selects a race, selects a horse, and chooses the predicted finishing rank from the dropdown menus.
       * The spectator clicks the "Gui du doan" button.
       * The system validates the prediction data and saves it to the database.
    2. **For managing prediction history:**
       * The system retrieves and displays the spectator's accumulated reward points and prediction history list (including Race, Horse, Predicted Rank, Result, and Actions).
       * If the race status is pending ("Cho ket qua"), the system allows the spectator to click "Sua" (Edit) or "Xoa" (Delete) to update their prediction.
       * If the race has ended, the actions are disabled ("Da khoa"), and the system updates the prediction status ("DUNG" / "SAI") along with the spectator's total reward points.

Screen layout:

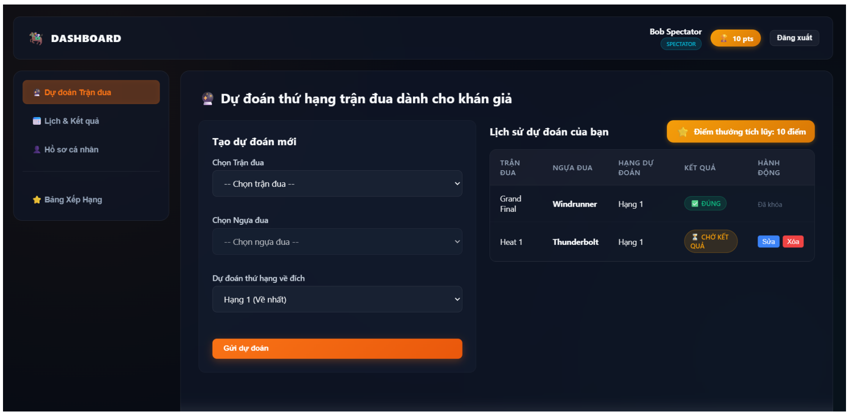


Figure 3.17: Spectator Prediction

View Schedule and Results

Role: Spectator

- **Function trigger:** The spectator navigates to the "Lich & Ket qua" (Schedule & Results) menu.
- **Function description:**
  - **Purpose:**
    1. Allow the spectator to view the list of upcoming race schedules.
    2. Allow the spectator to view the detailed results and rankings of completed races.
  - **Data processing:**
    1. **For upcoming race schedules ("Lich thi dau sap dien ra"):**
       * The system retrieves and displays scheduled races with status "SCHEDULED".
       * Information includes: Race name (e.g., Heat 1), referee name, date, time, distance, track condition, and the list of participating horses ("Danh sach tham gia") with their respective jockeys.
    2. **For completed race results ("Ket qua cac tran da ket thuc"):**
       * The system retrieves and displays finished races with status "HOAN THANH".
       * For each completed race (e.g., Grand Final), the spectator can expand to view the detailed ranking table including: Rank ("Hang"), Horse ("Ngua dua"), Jockey, Score ("Diem"), and Notes ("Ghi chu").

Screen layout:

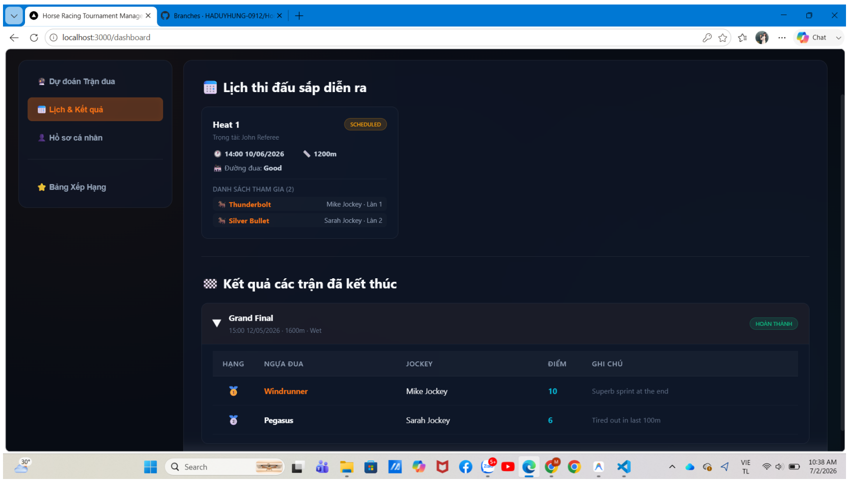


Figure 3.18: View Schedule and Results

Personal Profile

Role: Spectator

- **Function trigger:** The spectator navigates to the "Ho so ca nhan" (Personal Profile) menu.
- **Function description:**
  - **Purpose:**
    1. Allow the spectator to view and update their personal profile information.
    2. Display the spectator's prediction statistics, performance overview, and recent prediction history.
  - **Data processing:**
    1. **For updating profile information:**
       * The spectator can view and edit the following fields: Full Name ("Ho va Ten"), Email, Phone Number ("So dien thoai"), Avatar URL ("URL Anh dai dien"), Favorite Horse Breed ("Giong ngua yeu thich"), and Favorite Jockey ("Nai ngua yeu thich").
       * The spectator clicks the "Cap nhat thong tin" button.
       * The system validates the entered data and updates the profile details in the database.
    2. **For viewing statistics and history:**
       * The system retrieves and displays statistical insights ("Thong ke & Thanh tich") including: Current Rank ("Thu hang hien tai"), Total Predictions ("Tong so tran du doan"), Accuracy Rate ("Ty le chinh xac"), and Reward Points ("Diem thuong").
       * The system retrieves and lists the recent prediction history ("Lich su du doan gan day") with columns: Race Name ("Ten tran dua"), Selected Horse ("Ngua da chon"), Time ("Thoi gian"), Status ("Trang thai"), and Reward Points ("Diem thuong").

Screen layout:

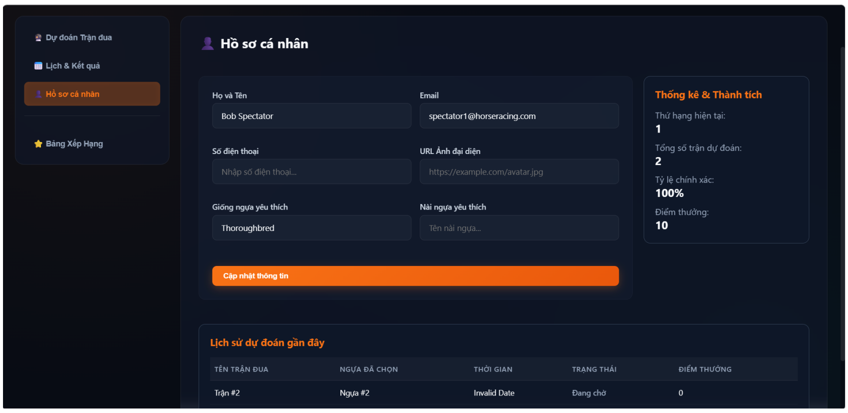


Figure 3.19: View Personal Profile

Leaderboard

Role: Spectator

- **Function trigger:** The spectator navigates to the "Bang Xep Hang" (Leaderboard) menu.
- **Function description:**
  - **Purpose:**
    1. Allow the spectator to view the official competition rankings of horses and jockeys, filtered by tournament.
    2. Allow the spectator to view the top 10 spectators with the highest prediction accuracy scores.
  - **Data processing:**
    1. **For Horse & Jockey Leaderboard ("Ngua & Jockey" tab):**
       * The spectator can filter rankings by tournament ("Loc theo giai dau") including options: All ("Tat ca"), Summer Championship 2026, or Spring Derby 2026.
       * The system retrieves and displays the Horse Ranking table ("Xep hang Ngua dua") including columns: Rank ("Hang"), Horse Name ("Ten ngua"), and Accumulated Points ("Diem tich luy").
       * The system retrieves and displays the Jockey Ranking table ("Xep hang Jockey") including columns: Rank ("Hang"), Jockey Name ("Ten jockey"), and Accumulated Points ("Diem tich luy").
    2. **For Top Spectators Leaderboard ("Khan gia xuat sac" tab):**
       * The system retrieves and displays the top 10 spectators with the highest prediction scores ("Top 10 Khan gia co diem du doan cao nhat").
       * The leaderboard table includes columns: Rank ("Hang"), Spectator Name ("Khan gia"), Favorite Horse Breed ("Giong ngua yeu thich"), and Accumulated Points ("Diem tich luy").

Screen layout:

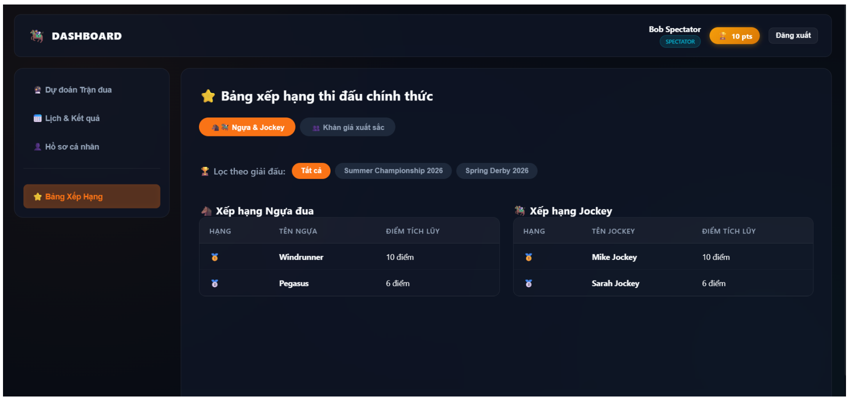


Figure 3.20: Official Competition Leaderboard for Horses and Jockeys

Screen layout:

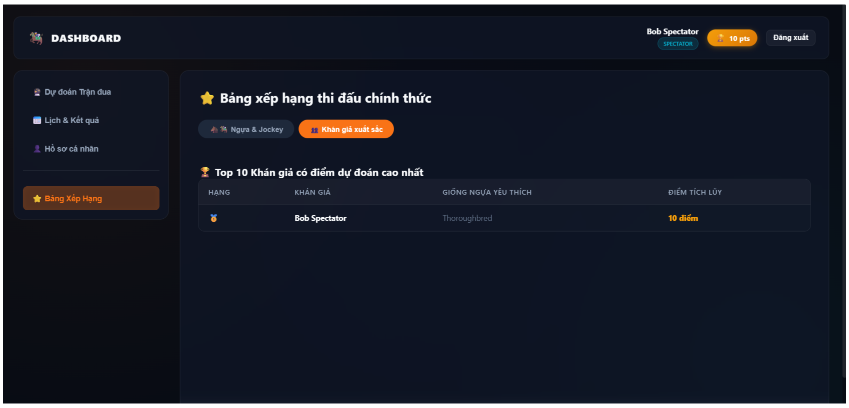


Figure 3.21: View Top 10 Spectators Prediction Leaderboard

Admin Login

Role: Admin

- **Function trigger:** The admin enters account credentials and selects the login button.
- **Function description:**
  - **Purpose:**
    1. Verify the administrator identity.
    2. Provide access to administrative functions.
  - **Data processing:**
    1. The admin enters username and password.
    2. The system validates the credentials.
    3. The system identifies the ADMIN role.
    4. The system creates an admin session.

- **Function details:**
  - Successful login redirects to the Admin Dashboard.
  - Invalid credentials show an error message.
  - Admin can access system management screens after login.

Screen layout:

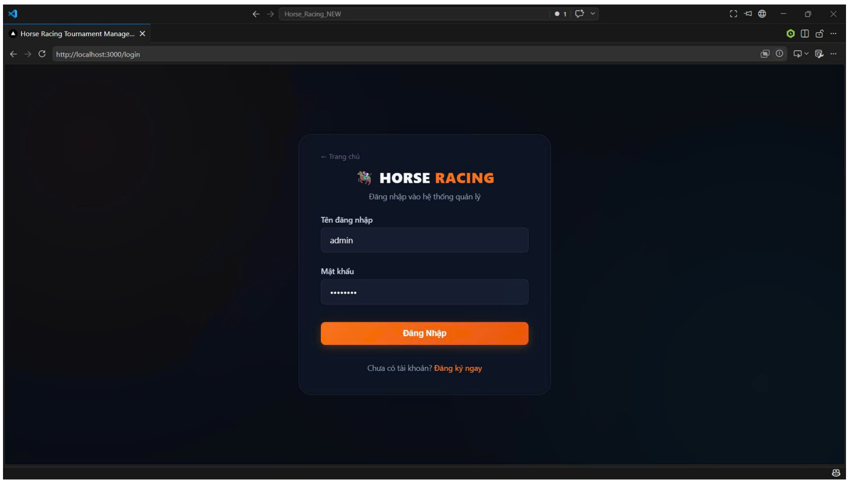


Figure 3.22: Admin Login

System Overview

Role: Admin

- **Function trigger:** Select "Tong quan he thong" from the admin menu.
- **Function description:**
  - **Purpose:**
    1. Provide a summary of system status.
    2. Display key metrics and role distributions.
  - **Data processing:**
    1. Retrieve user counts by role.
    2. Retrieve tournament and race statistics.
    3. Retrieve award and registration metrics.
- **Function details:**
  - Display total active users, tournaments, races, horses, and awards.
  - Show distribution of users by role.
  - Show tournament status and prediction statistics.

Screen layout:

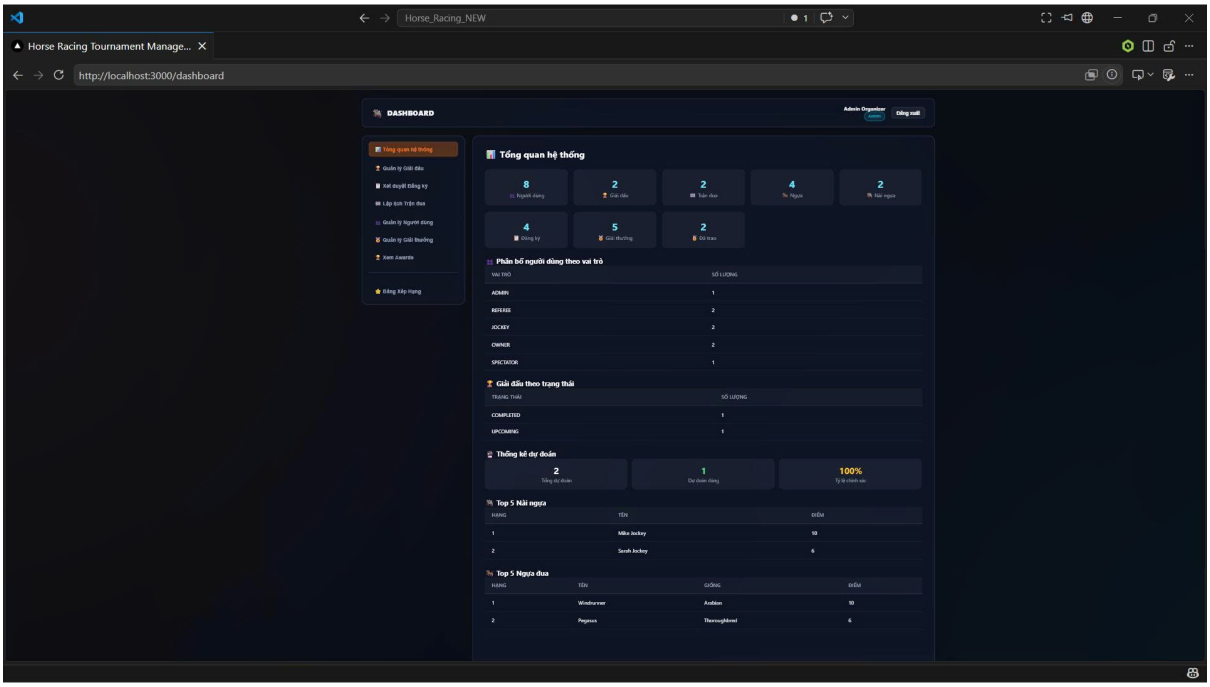


Figure 3.23: System Overview

Manage Tournaments

Role: Admin

- **Function trigger:** Select "Quan ly Giai dau".
- **Function description:**
  - **Purpose:**
    1. Create and manage tournament information.
    2. Add new rounds to tournaments.
  - **Data processing:**
    1. Enter tournament name, description, dates, and location.
    2. Save tournament details.
    3. Add competition rounds with sequence order.
- **Function details:**
  - Create new tournaments.
  - Edit or delete tournaments.
  - Manage the number and order of rounds.
  - Display current tournament list and status.

Screen layout:

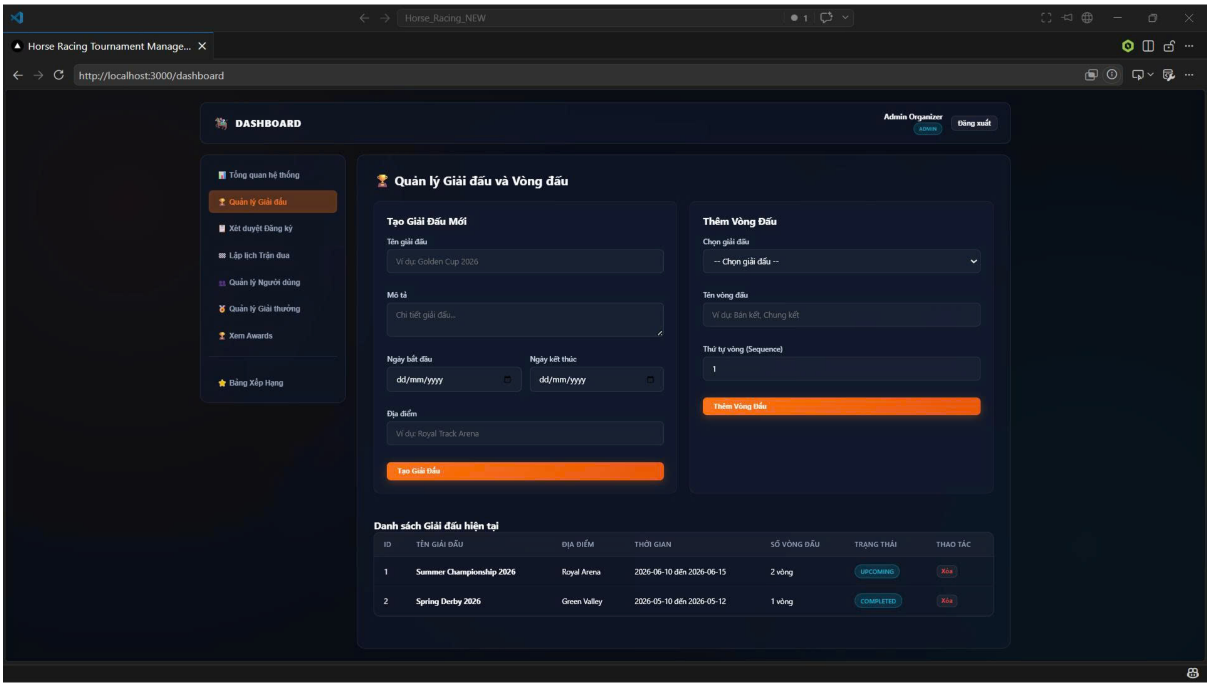


Figure 3.24: Manage Tournaments

Schedule Races

Role: Admin

- **Function trigger:** Select "Lap lich Tran dua".
- **Function description:**
  - **Purpose:**
    1. Schedule race heats and assign race details.
    2. Organize race lanes and pairs.
  - **Data processing:**
    1. Choose the tournament round.
    2. Enter race name, date, distance, and conditions.
    3. Select approved horse-jockey pairs.
    4. Assign lane numbers for each pair.
- **Function details:**
  - Create and manage races.
  - Assign horses and jockeys to lanes.
  - Display current race schedule and status.

Screen layout:

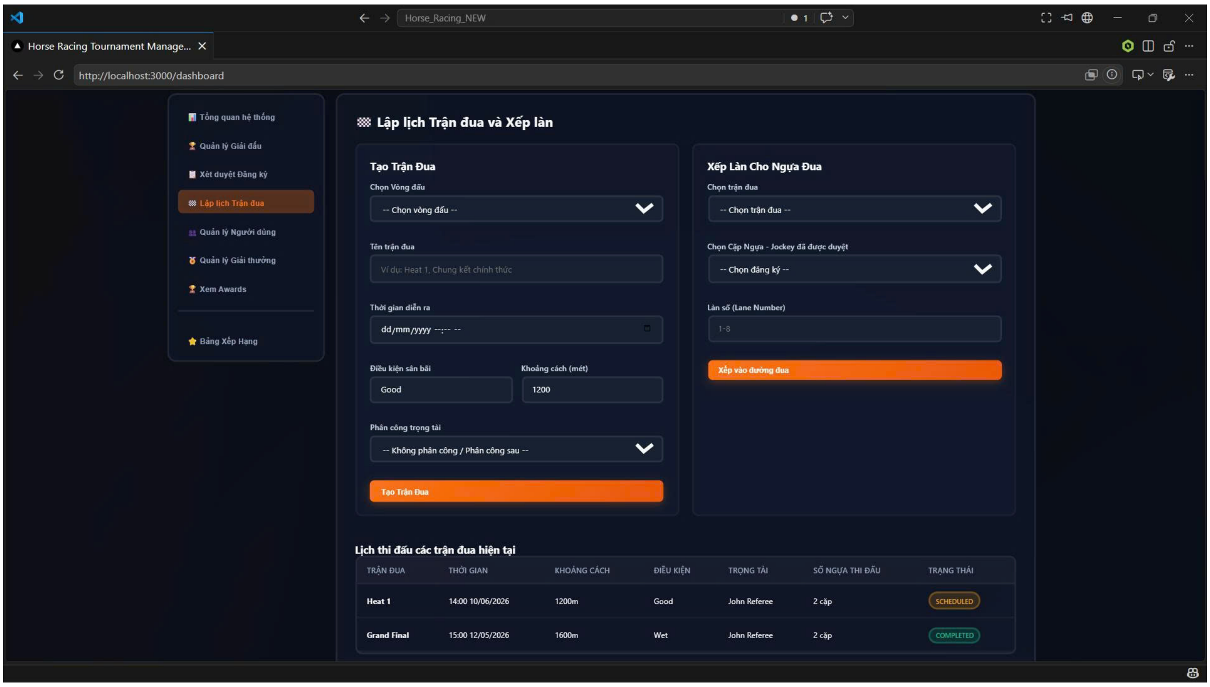


Figure 3.25: Schedule Races

Manage Rewards

Role: Admin

- **Function trigger:** Select "Quan ly Giai thuong".
- **Function description:**
  - **Purpose:**
    1. Define prizes for tournament positions.
    2. Manage award distributions.
  - **Data processing:**
    1. Select the tournament.
    2. Enter prize rank, title, value, and description.
    3. Save reward information.
- **Function details:**
  - Add new awards for tournaments.
  - Edit or delete existing awards.
  - Display award list and delivery status.

Screen layout:

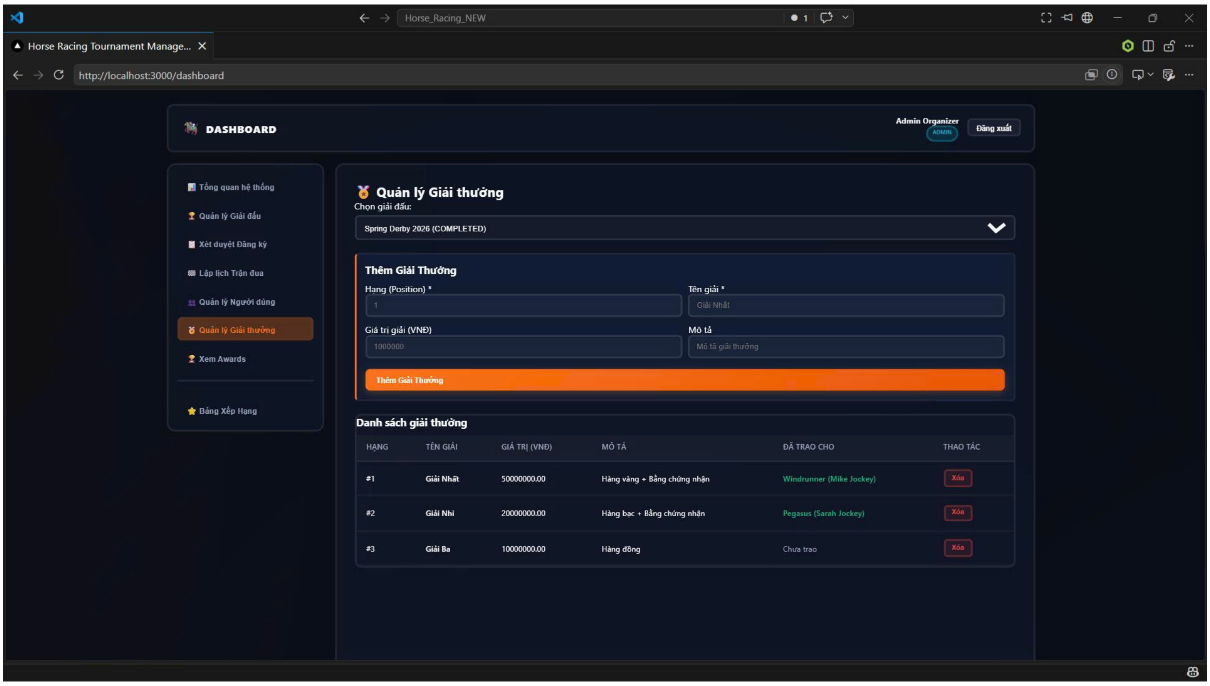


Figure 3.26: Manage Rewards

User Management

Role: Admin

- **Function trigger:** Select "Quan ly Nguoi dung".
- **Function description:**
  - **Purpose:**
    1. Manage system users and roles.
    2. Control access and account status.
  - **Data processing:**
    1. Retrieve user accounts.
    2. Display role and status information.
    3. Perform account lock or unlock actions.
- **Function details:**
  - Display user list with roles and statuses.
  - Lock or unlock user accounts.
  - Filter users by role.

Screen layout:

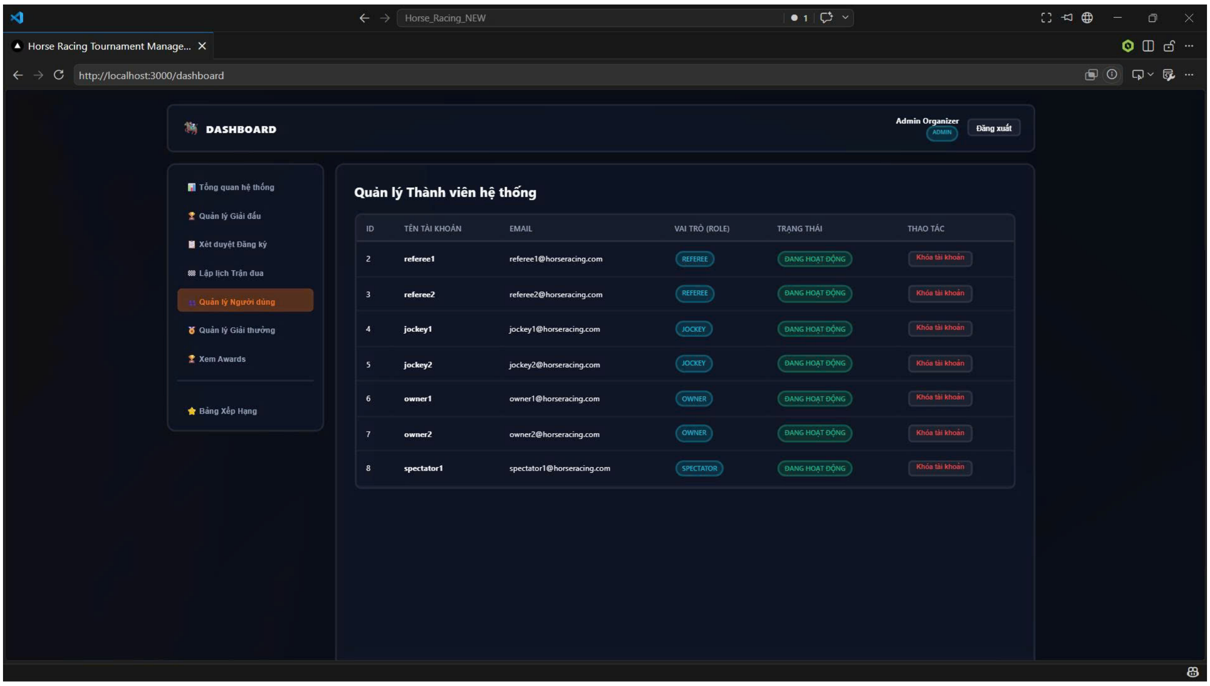


Figure 3.27: User Management

Confirm Registrations

Role: Admin

- **Function trigger:** Select "Xet duyet Dang ky".
- **Function description:**
  - **Purpose:**
    1. Review and confirm registration requests.
    2. Approve or reject tournament entries.
  - **Data processing:**
    1. Retrieve pending registrations.
    2. Display horse, jockey, and tournament details.
    3. Update registration approval status.
- **Function details:**
  - Show pending registration entries.
  - Approve or reject registrations.
  - Display current approval status.

Screen layout:

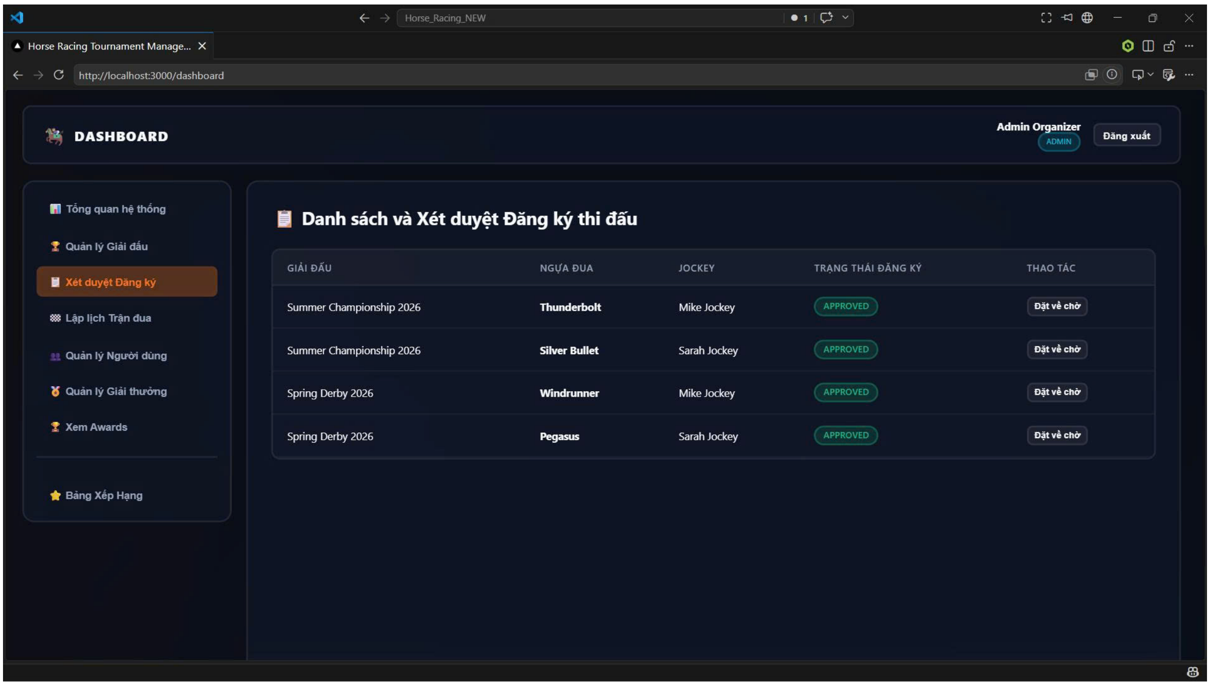


Figure 3.28: Confirm Registrations

Leaderboard

Role: Admin

- **Function trigger:** Select "Xem Awards" or "Bang Xep Hang".
- **Function description:**
  - **Purpose:**
    1. Display the official leaderboard.
    2. Review award winners and rankings.
  - **Data processing:**
    1. Retrieve ranking data for horses and jockeys.
    2. Retrieve award recipient details.
- **Function details:**
  - Display award and ranking summaries.
  - Show top positions by horse and jockey performance.

Screen layout:

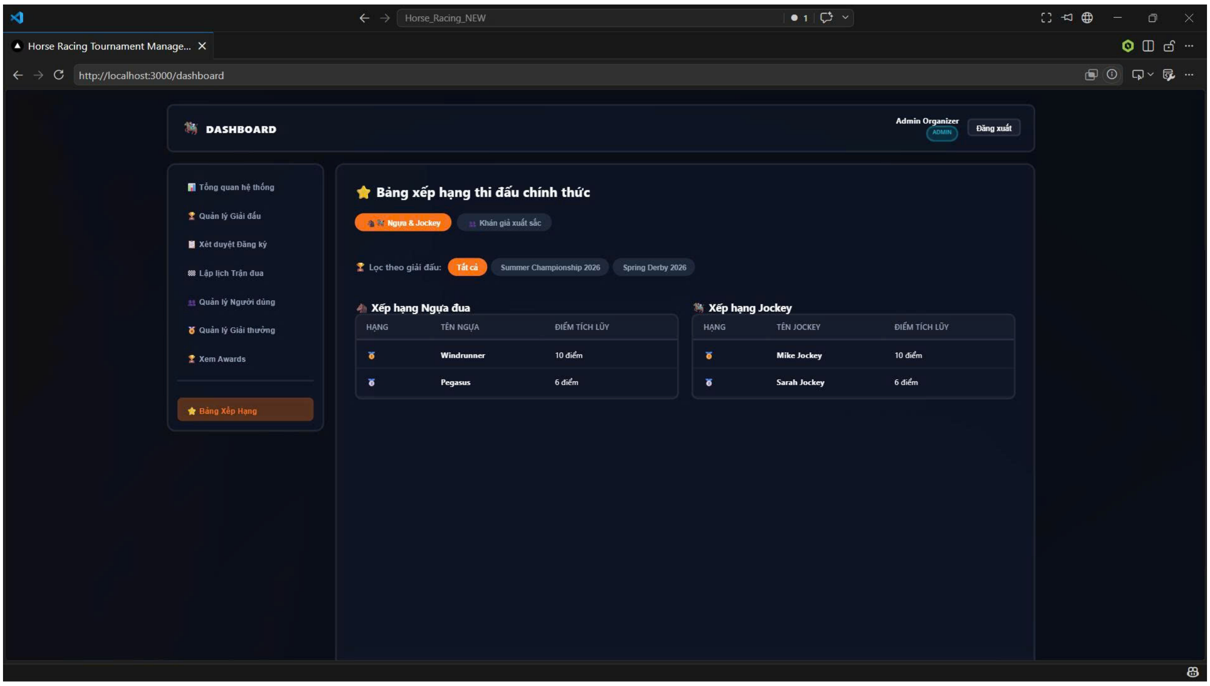


Figure 3.29: Leaderboard

3.4. Non-Functional Requirements

3.4.1 External Interfaces

To ensure effective communication between users and external systems, the following interfaces shall be supported.

User Interface

- The system shall provide a user-friendly and intuitive interface for Horse Owners, Jockeys, Referees, and Administrators.
- The graphical interface shall follow modern web design standards and support responsive layouts.
- Users shall be able to access the system through desktop computers, tablets, and mobile devices.
- Navigation menus and functions shall remain consistent throughout the system.

Authentication Interface

- The system shall support secure authentication using username and password.
- Role-based access control shall restrict system functions according to user roles.
- User sessions shall be managed securely to prevent unauthorized access.

Database Interface

- The application shall communicate with the database system to store and retrieve information.
- Data synchronization between the application server and database server shall be maintained.
- The database shall support CRUD operations for horses, tournaments, registrations, invitations, and users.

3.4.2 Quality Attributes

Usability

User-Friendly Interface

- The system shall provide a simple and intuitive interface.
- Functions such as tournament registration and jockey invitation shall require minimal user actions.
- The interface shall be easy to understand for both novice and experienced users.

Training and Familiarization Time

- Horse Owners and Jockeys shall require less than 30 minutes to learn the basic functions.
- Administrators shall require approximately 1–2 hours to understand management operations.

Usability Metrics

- Tournament registration shall be completed within 3 minutes.
- Error messages shall clearly guide users to correct invalid input.
- User feedback shall be collected to improve usability.

Reliability

Availability

- The system shall maintain at least 99% availability.
- Maintenance shall be scheduled during low-traffic periods.

Data Integrity

- Registration and tournament information shall be stored accurately.
- Duplicate registrations shall be prevented.

- Data consistency shall be maintained across all modules.

Error Handling

- The system shall display meaningful error messages.
- Critical system errors shall be logged for administrators.
- The system shall recover from failures with minimal data loss.

Performance

Response Time

- Normal user requests shall respond within 2 seconds.
- Under heavy load conditions, response time shall not exceed 5 seconds.

Concurrent Users

- The system shall support at least 500 concurrent users.
- Multiple users shall access the system simultaneously without significant performance degradation.

Database Performance

- Search operations shall complete within 3 seconds.
- Registration transactions shall be processed efficiently.

Security

- Passwords shall be encrypted before storage.
- Unauthorized access attempts shall be logged.
- Session timeout mechanisms shall automatically log out inactive users.
- Input validation shall prevent invalid data submission.

Maintainability

- The system shall follow a modular architecture.
- Source code shall comply with coding standards.
- New features shall be added with minimal impact on existing modules.
- API endpoints and database structures shall be documented.

Portability

- The system shall support Google Chrome, Microsoft Edge, and Mozilla Firefox.
- The application shall operate on both Windows and Linux servers.
- Users shall access the system without additional software installation.

3.5. Requirement Appendix

IV. Software Design Description

This chapter presents the software architecture, database design and detailed design.

4.1. Overall Design

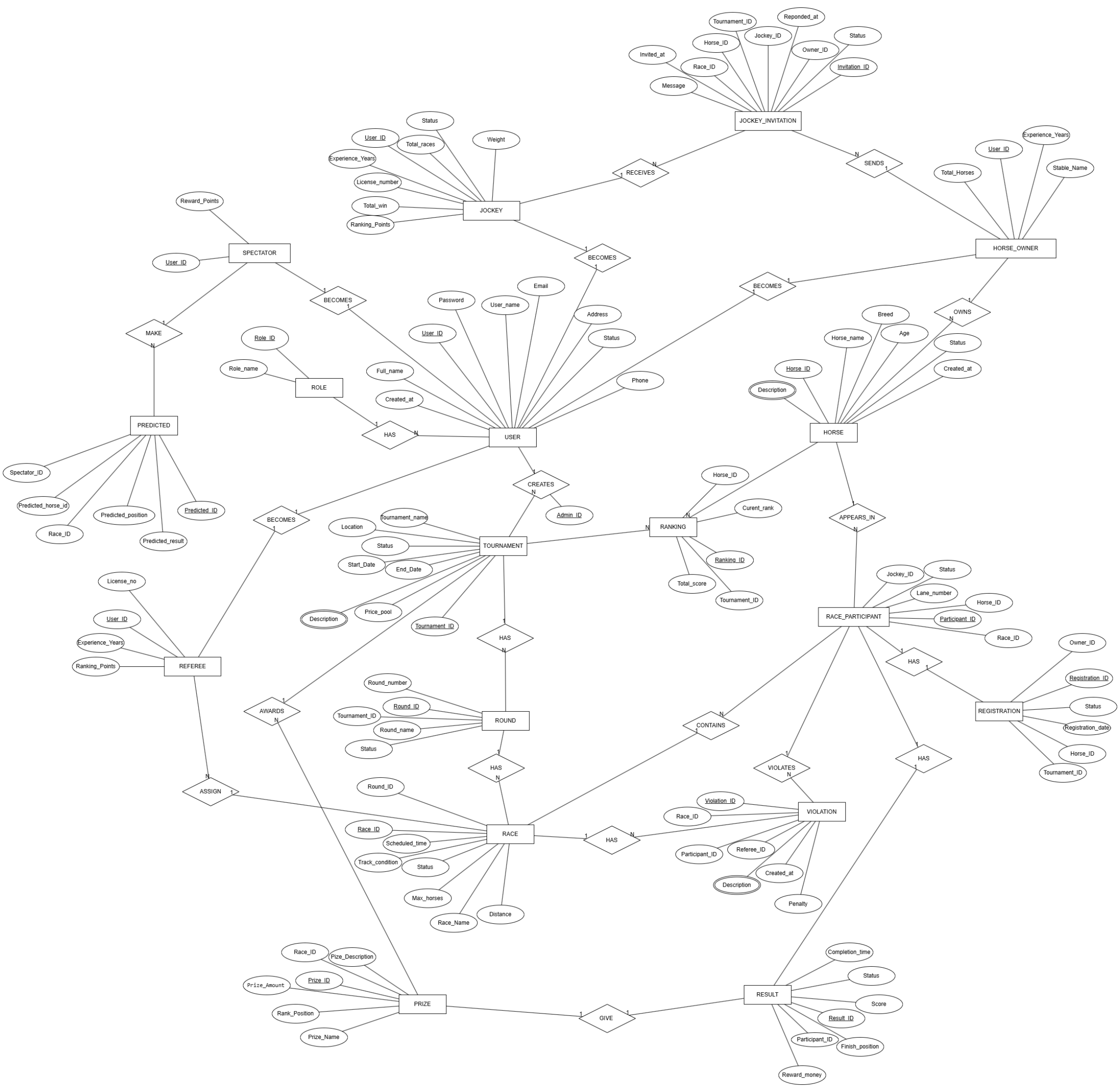


Figure 4.1: Entity Relationship Diagram

4.2. Detailed Design

4.2.1 Admin

4.2.2 Horse Owner Login and Registration

Sequence Diagram

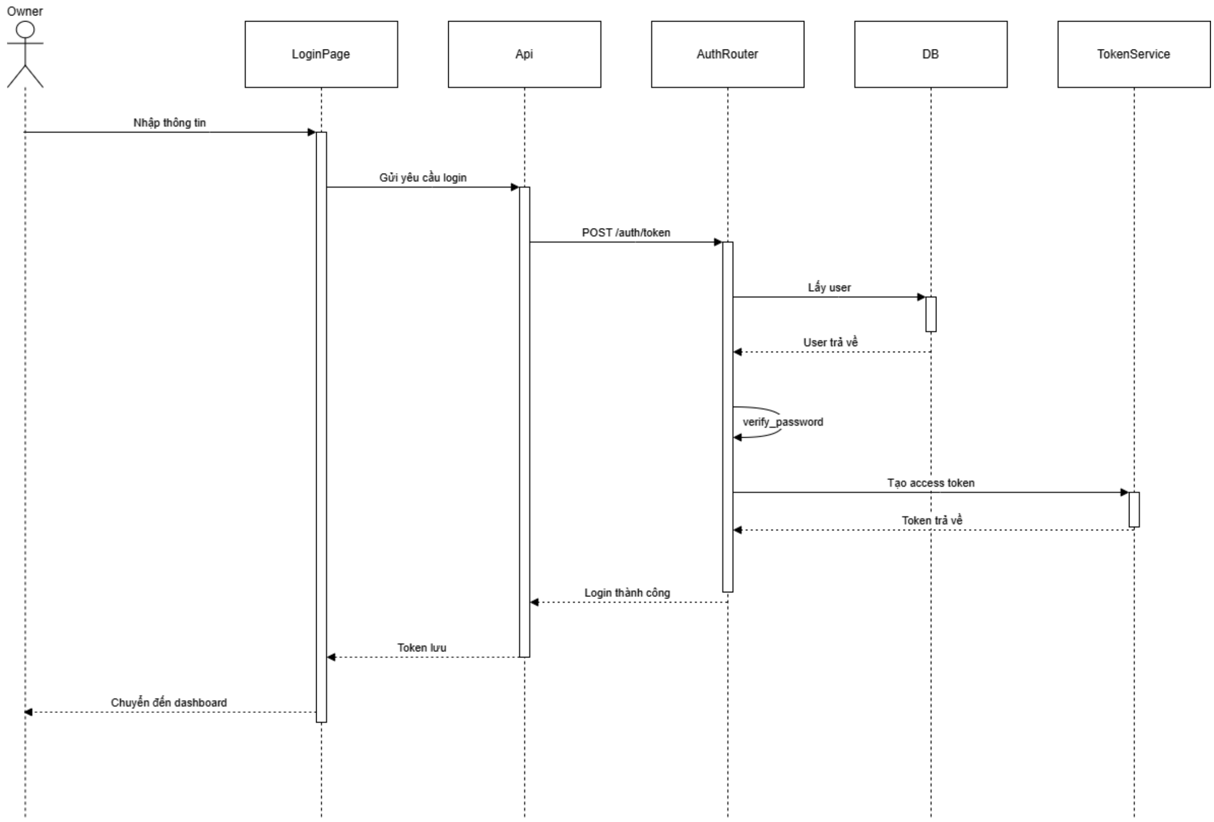


Figure 4.2: Horse Owner Login Sequence Diagram

Class Diagram

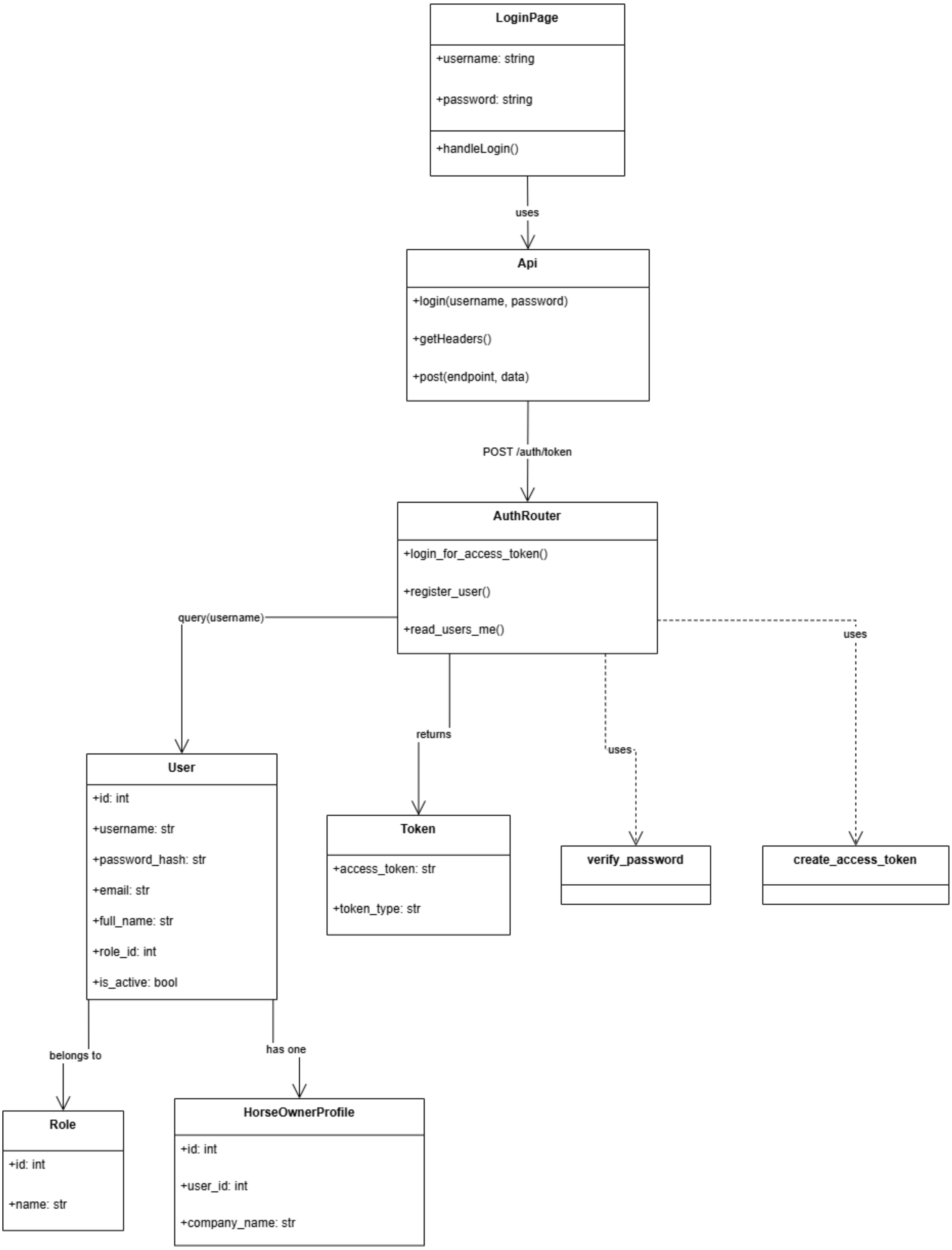


Figure 4.3: Horse Owner Login Class Diagram

Class Specification

Class	Attributes / Methods	Type	Description
LoginPage	username, password, handleLogin()	state / action	UI component that collects login credentials and calls the API.
Api	login(), post(), getHeaders()	service / helper	Calls `POST /auth/token`, stores JWT, and prepares request headers.
AuthRouter	login_for_access_token(), read_users_me()	controller / route	Validates username/password and issues bearer tokens.
User	id, username, password_hash, email, full_name, role_id, is_active	entity	Database entity representing an application user.
Role	id, name	entity	Defines user roles, including `OWNER`.
Token	access_token, token_type	DTO	Response payload returned after successful login.
HorseOwner Profile	id, user_id, company_name	entity	Owner-specific profile linked to the User entity.

Table 4.1: Horse Owner Login class specification

4.2.3 Jockey

4.2.4 Spectator

4.2.5 Referee